

STAKEHOLDER 32

School leadership, university leadership, and building owners

Practical, evidence-aware stakeholder question catalogue

80

DECISION-FACING QUESTIONS

8

LEADERSHIP DOMAINS

6

EVIDENCE FIELDS PER RECORD

1

Mission, strategy and stakeholder outcomes

2

Capacity, access, experience and service demand

3

Operations, teaching, research and workplace performance

4

Risk, safety, compliance and assurance

5

Finance, assets, projects and long-term value

6

Energy, carbon, inclusion and social value

7

Resilience, disruption and business continuity

8

Governance, accountability, scenarios and performance reporting

Scope and operating stance

The service frames evidence for accountable leaders; it does not become the decision owner.

Purpose before evidence

Each card begins with a real leadership or ownership decision, then identifies the minimum evidence and accountable action.

Proposed, not presumed

Sensors, integrations and analytics remain proposed until deployment, mapping, commissioning, permission and current health are independently verified.

Current authority takes precedence

Current records and competent professional statements take precedence within their scope. Emergency command and authorised instructions outrank analytics.

Historical plan only

The May 2021 floor plan supports historical spatial orientation only. It cannot establish current room use, access, services, assets, hazards or closures.

Decision support, not authority

The service does not approve investment, allocate staff, accept risk, certify compliance, control building systems or replace accountable governance.

Honest safe failure

When a required source, permission, mapping, health check or minimum evidence condition is absent, the answer is insufficient evidence with the gap and owner named.

Question-card field guide

Read all six fields together. A plausible question without sufficient evidence remains unanswered.

1**Stakeholder question**

A natural question tied to a specific strategic, operational, financial, safety or governance decision.

2**Sensors and telemetry**

Proposed, privacy-preserving evidence with asset and location scope, timing, aggregation, acceptance checks and known blind spots.

3**Authoritative supporting data**

Current owners, records, precedence, permissions, provenance, effective-observed-retrieved times, staleness and unresolved conflict.

4**Analysis required**

Operation type, decision interval, minimum evidence, method, quantified uncertainty and a named safe-failure route.

5**Answer boundary**

What the service must not infer, disclose, promise, certify, authorise or control.

6**Why this is useful**

The concrete decision, commission, escalation, prioritisation or evidence action available to accountable leaders.

READING SEQUENCE

Question -> proposed evidence -> current authority -> analytic conditions -> boundary -> accountable action

Evidence, timing and permission gates

Every gate is claim-specific. A healthy system cannot make an unsupported answer safe.

01 Decision and authority Name the decision, accountable owner, recipients, horizon and permitted use before evidence access.	02 Minimum necessary access Use the least data and coarsest aggregation that can answer the approved question.	03 No identification or tracking Do not identify, follow or reconstruct attendance, journeys, routines or behaviour.
04 No content inspection Do not inspect speech, images, video, messages, files, teaching, research or network content.	05 Current mapping required Verify current room, route, asset, meter and owner mappings; historical spatial labels are not current evidence.	06 Three times and one cut-off Retain effective, observed and retrieved times and freeze the evidence cut-off for the decision.
07 Aggregation and suppression Release the largest useful aggregation and suppress small, unsafe or re-identifiable combinations.	08 Provenance and live health Carry source, owner, permissions, mapping, completeness, calibration, transformations and current health.	09 Conflict remains visible Apply precedence only within scope; preserve material disagreement for the authorised resolver.
10 No personal performance inference Presence, comfort and service use do not establish individual effort, quality, intent or outcome.	11 No automatic authority Analytics do not control plant, doors, alarms, bookings, communications, staffing or funding.	12 Professional and emergency precedence Competent advice, current on-scene instruction and authorised command take precedence.

Catalogue map and answer discipline

Eight categories, ten questions each, two question cards per page.

1 Mission, strategy and stakeholder outcomes LB-001 to LB-010 pages 6-10	2 Capacity, access, experience and service demand LB-011 to LB-020 pages 11-15
3 Operations, teaching, research and workplace performance LB-021 to LB-030 pages 16-20	4 Risk, safety, compliance and assurance LB-031 to LB-040 pages 21-25
5 Finance, assets, projects and long-term value LB-041 to LB-050 pages 26-30	6 Energy, carbon, inclusion and social value LB-051 to LB-060 pages 31-35
7 Resilience, disruption and business continuity LB-061 to LB-070 pages 36-40	8 Governance, accountability, scenarios and performance reporting LB-071 to LB-080 pages 41-45

ANSWER DISCIPLINE

1 Frame Decision, authority, horizon and duties	2 Verify Mapping, permission, health and cut-off	3 Test Minimum evidence, method and uncertainty	4 Bound Fact, estimate, scenario and unknown	5 Route Useful action and accountable owner
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LB-001

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which building capabilities must be protected first to deliver the School's approved teaching, research, civic and workforce priorities, and where is the current evidence of shortfall strongest?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: overhead thermal entrance counters, ceiling occupancy-state sensors, BMS plant availability, room-service alarms and representative temperature, CO2 and lighting telemetry. Spatial coverage: only currently mapped teaching, research-support, workplace, public-facing and shared-service zones named in the outcome map. Observation and aggregation: events or 1-5 minute samples, summarised daily, weekly and by representative teaching period. Decision-use gate: at least 90% valid data in every cited zone across six representative weeks, with calibration and clock checks recorded. Known limits: presence and conditions are service proxies, not evidence of learning, research quality or institutional outcome achievement. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School executive and strategy leads, University planning, Estates service owners, timetabling and space management. Required records: approved strategy and outcome definitions, current operating standards, timetable, bookings, outage and service records. Source precedence: current approved outcomes, schedules and owner-confirmed service status outrank inferred use and the May 2021 floor plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: a superseded strategy, changed timetable or telemetry beyond its owner-set health window is stale. Unresolved conflicts: local and University priorities, scheduled and actual availability, and differently defined outcome measures; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: outcome-to-service dependency mapping and exception ranking. Decision interval: monthly and at each term boundary. Minimum evidence: current outcome owners, a signed service map, six representative weeks and complete status for every claimed dependency. Method: link each approved outcome to observable building capabilities, compare required and observed service, and rank only material, repeated gaps. Quantified uncertainty: show coverage by zone, ranges around rates, confidence class and alternative explanations; do not claim causality; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the School strategy owner and relevant Estates service owner.

ANSWER BOUNDARY

The service must not infer achievement, learning, research quality, staff effectiveness or stakeholder satisfaction from presence or environmental conditions. It must not disclose individual or small-group activity, restricted research locations or security-sensitive service details. It must not promise or certify mission success, benefit causation or completeness of the evidence; it must not authorise or control priorities, budgets, staffing, room allocation or building services. The answer is a dependency view, not a judgement on institutional performance.

WHY THIS IS USEFUL

School and Estates leaders can protect the capabilities most critical to the approved mission and assign named owners to evidenced shortfalls.

LB-002

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which verified building barrier is causing the greatest avoidable loss of an approved stakeholder outcome, and who needs to own the next action?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: lift and powered-door status, door-contact and route-availability signals, room and amenity availability, plus optional anonymous issue-category feedback. Spatial coverage: principal accessible journeys and destination services verified in the current access map; no demographic sensing or person-level route linkage. Observation and aggregation: status changes as events, daily issue counts and weekly or term summaries with small-count suppression. Decision-use gate: verified asset-to-route mapping, event delivery above 95%, duplicate feedback removal and at least two independent evidence types for a priority claim. Known limits: silent barriers, non-use, inaccessible feedback channels and temporary assistance arrangements may not appear in telemetry. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accessibility and inclusion teams, affected service owners, student and staff experience leads, complaints and adjustment-support services. Required records: current access assessments, Personal Emergency Evacuation Plan governance, service tickets, approved access guides and aggregate feedback. Source precedence: lived experience and competent current accessibility assessment outrank sensor-based assumptions or historical room labels. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: access guides and route maps become stale immediately after layout, closure, works or service changes. Unresolved conflicts: reported experience, asset status, temporary arrangements and different owners' definitions of a resolved barrier; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: barrier triangulation, severity classification and accountable-owner routing. Decision interval: termly, with urgent exceptions examined when reported. Minimum evidence: current route mapping, verified feature status, safe aggregation, an affected-service owner and corroboration for any prioritised pattern. Method: group issues by place, service and time; separate frequency, consequence, recurrence and evidence quality; retain unresolved lived-experience evidence. Quantified uncertainty: show response bias, suppressed cells, blind spots and a confidence class rather than a single prevalence estimate; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accessibility lead, relevant service owner and affected stakeholder representatives.

ANSWER BOUNDARY

The service must not infer disability, protected characteristics, health status, personal needs, intent or legitimacy of use. It must not disclose case details, free-text comments, individual journeys or small groups. It must not promise or certify equality compliance, absence of barriers or adequacy of a reasonable adjustment; it must not authorise or control adjustments, route restrictions, service priorities or capital work. Competent assessment and direct engagement remain necessary.

WHY THIS IS USEFUL

The accountable executive can select the first barrier for action, commission direct stakeholder engagement and track whether the dependency is cleared.

LB-003

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which spaces have materially drifted from their current approved purpose, capacity or service requirement enough to justify an allocation decision?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: ceiling PIR or mmWave occupancy state, anonymous doorway counts, booking-panel state and non-content equipment run-state. Spatial coverage: rooms and shared zones only after current identifiers, ownership, capacity and function are confirmed. Observation and aggregation: 5-15 minute observations during operating hours, aggregated by week, teaching period and term. Decision-use gate: at least four representative weeks, 85% valid occupancy-state coverage and explicit treatment of specialist, confidential and contingency use. Known limits: occupancy cannot reveal purpose, preparation, quality or legitimate low-frequency specialist activity. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School space lead, central space management, timetabling, room-booking owners and asset information management. Required records: current space inventory, approved allocation, capacity, functional brief, bookings, timetable and handover records. Source precedence: current approved designation and owner-confirmed exceptions outrank inferred activity and the May 2021 floor plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: room mappings are stale after renaming, subdivision, decant, project handover or change of owner until revalidated. Unresolved conflicts: local allocation, central inventory, bookings, temporary use and sensor-to-room mapping; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: planned-versus-observed space-use classification. Decision interval: termly and before allocation or repurposing gateways. Minimum evidence: current ownership and purpose, verified capacity, four representative weeks and explicit rules for specialist and resilience space. Method: compare scheduled, booked and anonymous presence patterns; flag sustained mismatch, under-supply and contested evidence without assigning motive. Quantified uncertainty: report room-level coverage, classification confidence and cases where presence is not an interpretable proxy for purpose; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the School space lead and central space-management owner.

ANSWER BOUNDARY

The service must not infer the content, legitimacy, productivity or quality of teaching, research or work from occupancy. It must not disclose who used a room, restricted project activity or small-group patterns. It must not promise or certify unauthorised use, strategic misalignment or a requirement to surrender space; it must not authorise or control allocation, booking cancellation, access or repurposing. Low observed use may be justified by specialist, confidential, accessibility or resilience requirements.

WHY THIS IS USEFUL

Space owners can retain, share, reschedule or propose repurposing with a transparent record of current purpose, demand and legitimate exceptions.

LB-004

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Within the available resource envelope, which building intervention gives the most defensible gain across mission outcomes rather than a narrow local benefit?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: option-specific aggregate demand, queue, availability, fault, comfort, energy and amenity-use indicators. Spatial coverage: only assets and locations in the defined options, with the same baseline boundary and service hours. Observation and aggregation: 1-15 minute operational samples as appropriate, monthly baselines and option-level summaries. Decision-use gate: comparable baselines, documented meter and counter health, known missingness below 10% for each scored measure and no unsupported benefit proxy. Known limits: hidden demand, unequal coverage, non-sensor benefits, confounding works and uncertain cost attribution. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accountable budget holder, service owners, finance partner, strategy lead, accessibility, sustainability and risk specialists. Required records: approved minimum standards, option scope, current cost ranges, service levels, risk controls, stakeholder evidence and delivery dependencies. Source precedence: legal, safety, access and contractual minimums constrain options before discretionary benefit ranking. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: costs are stale after scope, market, inflation or programme change; service baselines are stale after material operational change. Unresolved conflicts: cost, risk, service, carbon, inclusion and stakeholder evidence, including non-monetisable effects; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: multi-criteria option appraisal with constraint screening. Decision interval: quarterly and at each funding gateway. Minimum evidence: verified baseline, named benefit owner, cost range, delivery dependencies and evidence for service, risk and inclusion effects. Method: remove infeasible options, score remaining options against published criteria, test alternative weights and present dominated options separately. Quantified uncertainty: show ranges, sensitivity to weights, unquantified benefits and the share of tested cases in which rankings change; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable budget holder and each relevant professional owner.

ANSWER BOUNDARY

The service must not infer that the highest analytic score is the preferred institutional choice. It must not disclose commercially restricted estimates, personal feedback or security-sensitive weaknesses beyond authorised roles. It must not promise or certify affordability, deliverability, value for money or benefit realisation; it must not authorise or control funding, procurement, staffing, scope or service change. Mandatory duties and professional advice remain decision constraints, not optional scoring dimensions.

WHY THIS IS USEFUL

The budget holder can shortlist an intervention whose mission benefit remains credible across alternative weights and cost ranges.

LB-005

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What can leadership responsibly claim about the building's readiness to prospective students, staff, partners, funders and visitors, and what must remain qualified?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: aggregate service uptime, access-feature status, environmental compliance indicators, verified incident counts and optional anonymous experience categories. Spatial coverage: public-facing, teaching, workplace and partner-use areas explicitly linked to the claim; no person-level conversion or retention tracking. Observation and aggregation: monthly and annual trend summaries with event dates and campaign or term context. Decision-use gate: at least two independent sources, a stable definition for 12 months and suppression of small cohorts. Known limits: selection bias, external causes, marketing exposure and changing expectations prevent causal attribution. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School and University communications, admissions, HR, partnership leads, service owners, complaints and reputation-risk owners. Required records: approved claims, current access and service statements, aggregate survey results, complaints, incident records and recruitment or retention statistics. Source precedence: verified current service facts and approved communications policy outrank anecdote, telemetry-only interpretation or historical descriptions. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: public claims are stale when the underlying service, evidence period or approval changes. Unresolved conflicts: operational evidence, stakeholder reports, campaign narratives and different definitions of reputation or retention impact; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: claim-evidence matching and contribution analysis. Decision interval: before publication and at least annually. Minimum evidence: a precisely worded claim, defined population and period, two evidence types, relevant comparator and named claim owner. Method: separate directly observed service facts from stakeholder perception and wider institutional outcomes; test whether plausible alternatives remain. Quantified uncertainty: state sample size, response bias, confidence interval and a claim-strength grade from descriptive to supported association; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the communications claim owner and relevant service or experience owner.

ANSWER BOUNDARY

The service must not infer why a person applied, left, stayed or formed an opinion. It must not disclose individual applicant, employee, partner or complainant information. It must not promise or certify causal effect on recruitment, retention or reputation, or a universally positive experience; it must not authorise or control public messaging, corrective action, staffing or investment. Only wording supported at the stated evidence strength should be presented.

WHY THIS IS USEFUL

Leadership can approve accurate prospectus, partnership and funding statements while clearly qualifying evidence gaps and time limits.

LB-006

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Where are teaching, research, workplace and public-facing priorities competing for the same room, route or service, and which trade-off needs an accountable decision?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous entrance and zone counts, room occupancy state, booking-panel status, lift and amenity availability, and service-load telemetry. Spatial coverage: shared resources and interfaces named in the current dependency map, not individual rooms with confidential activity unless safely aggregated. Observation and aggregation: 5 minute to hourly data, summarised by day type, teaching week and event window. Decision-use gate: clock alignment within one sampling interval, at least six representative weeks and valid data above 90% during claimed peaks. Known limits: bookings may not become use, informal demand is incomplete and occupancy cannot identify activity type. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School operations, timetabling, research facilities, events, workplace, security and Estates service owners. Required records: approved schedules, bookings, event plans, service capacities, access constraints and owner-confirmed priority rules. Source precedence: current approved commitments and safety or access constraints outrank inferred demand or first-come telemetry. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: schedules and capacities are stale after approved change, closure, decant or service derating. Unresolved conflicts: competing owner priorities, central and local bookings, service capacity and temporary operating constraints; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: time-space demand overlap and constrained-resource analysis. Decision interval: weekly in peak periods and before major scheduling cycles. Minimum evidence: current commitments, verified resource capacities, six representative weeks and named priority or escalation rules. Method: build demand envelopes by activity class, identify overlaps exceeding usable capacity and test staggering, relocation and service alternatives. Quantified uncertainty: show demand ranges, unobserved informal use, capacity confidence and scenarios where the conflict disappears; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the School operations lead and owners of each competing service.

ANSWER BOUNDARY

The service must not infer the value, legitimacy or priority of an activity from its occupancy or schedule. It must not disclose individual attendance, research content, event participants or confidential bookings. It must not promise or certify that a conflict requires cancellation, relocation or reduced service; it must not authorise or control timetables, bookings, access, service priorities or staffing. Priority remains an accountable governance choice informed by explicit duties and commitments.

WHY THIS IS USEFUL

Leaders can govern a specific trade-off, protect non-negotiable activities and set a monitored allocation or service rule.

LB-007

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What baseline, acceptance criteria and follow-up evidence must be agreed before a proposed building investment is approved?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: only option-relevant aggregate service, availability, demand, environmental, energy and access-feature indicators with baseline health metadata. Spatial coverage: the exact spaces, assets, services and affected groups stated in the approved benefits map. Observation and aggregation: baseline and post-handover periods matched by season and academic context, with monthly reporting. Decision-use gate: measures have named owners, stable definitions, verified mapping and sufficient pre-intervention history to estimate normal variation. Known limits: many strategic, inclusion and experience benefits require direct engagement or professional evidence rather than sensors. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: business-case owner, project sponsor, School outcome owners, finance, Estates, sustainability, accessibility and stakeholder-representation leads. Required records: approved business case, benefits map, design requirements, acceptance criteria, baseline records and post-occupancy evaluation plan. Source precedence: approved duties and acceptance criteria precede convenience metrics; current owner-approved benefit definitions outrank later analytic substitutions. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: a measure is stale after scope, user group, service or definition changes unless formally rebaselined. Unresolved conflicts: sponsor, service and stakeholder interpretations of success, metric boundaries and attributable benefit; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: benefit-measure design and evaluability assessment. Decision interval: before approval, at handover and at planned benefits assessments. Minimum evidence: named benefit owners, baseline window, target or decision threshold, counterfactual rationale, data route and equity check for every material benefit. Method: test whether each measure is observable, attributable enough for its claim, resistant to gaming and paired with qualitative or professional evidence where needed. Quantified uncertainty: show baseline variance, detectable effect size, confounders and benefits that cannot be quantified; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project sponsor and accountable benefit owners.

ANSWER BOUNDARY

The service must not infer learning, research quality, wellbeing, inclusion or productivity from a convenient proxy. It must not disclose individual-level outcomes, protected characteristics or small affected groups. It must not promise or certify that meeting a target proves overall success or causation; it must not authorise or control project approval, target setting, benefit ownership or acceptance. A measure can inform a decision only within its declared scope and evidence strength.

WHY THIS IS USEFUL

The sponsor can set measurable acceptance criteria, a baseline window, evidence ownership and a post-handover decision point before committing funds.

LB-008

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which opening, staffed-support or specialist-service hours are demonstrably misaligned with legitimate mission demand across the year?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: overhead thermal entrance counts, room occupancy state, help-point or service-desk aggregate transactions, lift and amenity availability, and security-alarm arming state. Spatial coverage: entrances and services included in the proposed hours decision, with after-hours zones separately mapped. Observation and aggregation: 5-15 minute data, aggregated by hour, day type, assessment period, vacation and event window. Decision-use gate: at least one full academic cycle where available, 90% valid peak-period coverage and known closure or exceptional-event flags. Known limits: counts do not reveal purpose, unmet demand, service quality or the staffing and safeguarding effort required. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School operations, security, library or service owners, HR, finance, accessibility, timetabling and events leads. Required records: approved opening policy, staffing and lone-working controls, schedules, bookings, incident logs, service standards and cost records. Source precedence: safety, labour, access and safeguarding requirements constrain hours before demand optimisation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: hours, staffing assumptions and demand history are stale after programme, timetable, transport or service changes. Unresolved conflicts: observed presence, booked demand, staff capacity, cost, risk and stakeholder requests; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: service-window demand and marginal-cost analysis. Decision interval: annually and before permanent or trial hour changes. Minimum evidence: representative demand by season, current control requirements, incremental cost range, service owner and a defined trial success criterion. Method: estimate demand envelopes and unmet demand, compare current and candidate windows, and run a time-limited trial where evidence is weak. Quantified uncertainty: show confidence intervals, exceptional periods, hidden demand and the sensitivity of value to staffing and security assumptions; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable service owner, security lead and budget holder.

ANSWER BOUNDARY

The service must not infer why a person is present, whether attendance is necessary or whether a service should be staffed. It must not disclose individual entry records, lone-worker identities or small-group routines. It must not promise or certify safety, affordability, equality or service quality of extended hours; it must not authorise or control opening, closure, staffing, access permissions or security controls. Any change requires current risk assessment and accountable approval.

WHY THIS IS USEFUL

Service owners can retain, extend, reduce or pilot operating hours with staffing, security, energy and equity implications visible.

LB-009

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Has a completed building change delivered the benefits promised to the named services and stakeholder groups, or only changed activity patterns?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: project-specific aggregate availability, occupancy context, environmental, energy, queue, fault and access-feature indicators. Spatial coverage: only the intervention area and agreed comparison areas, with unchanged sensor boundaries or documented remapping. Observation and aggregation: matched baseline and post-handover windows, monthly summaries and milestone snapshots. Decision-use gate: commissioned sensors, at least eight comparable weeks before and after, stable definitions and explicit treatment of defects and seasonal variation. Known limits: benefit take-up, distribution and lived experience may be partly or wholly unobservable in telemetry. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: project sponsor, business-case owner, School and service benefit owners, finance, accessibility, sustainability and user-representation leads. Required records: approved benefits map, scope, handover and defect records, acceptance tests, baseline, stakeholder evidence and benefit assessment decisions. Source precedence: approved benefit definitions and current competent acceptance evidence outrank post hoc metrics or historical spatial assumptions. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: baseline or target is stale after material scope, service, population or operating-model change unless reappraised. Unresolved conflicts: business-case promise, delivered scope, service data, affected-group evidence and responsibility for unrealised benefits; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: benefit realisation and distributional assessment. Decision interval: at handover, end of defects period and agreed annual assessments. Minimum evidence: traceable benefit owner, comparable baseline, verified delivered scope, affected-service evidence and a distribution check. Method: compare actual conditions with baseline and target, stratify only at safe aggregate levels, and separate non-delivery, non-use and unmeasured benefit. Quantified uncertainty: show effect ranges, missing groups, confounders and whether the evidence supports description, association or a stronger claim; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project sponsor and each accountable benefit owner.

ANSWER BOUNDARY

The service must not infer individual benefit, satisfaction, learning, productivity or research success from aggregate use. It must not disclose personal outcomes, protected characteristics, case details or commercially restricted defects. It must not promise or certify causation, equitable benefit, project success or completion of all obligations; it must not authorise or control remedial work, benefit acceptance, payment, scope or future investment. Unmet or unmeasured benefits remain visible rather than being treated as zero.

WHY THIS IS USEFUL

The sponsor can accept benefits, require corrective action, extend measurement or stop unsupported claims against the original case.

LB-010

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which strategic building decision is not yet evidence-ready, and what is the smallest proportionate evidence action that would unblock it?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: source-health, mapping, coverage and timestamp telemetry for the specific meters, counters, BMS points or service-status feeds cited by the decision. Spatial coverage: only evidence routes in the active decision pack, not a general data-quality score for the whole building. Observation and aggregation: continuous health events and a snapshot at each decision cut-off. Decision-use gate: every cited source has an owner, current mapping, permission, completeness estimate and last successful validation. Known limits: a healthy feed can still be irrelevant, biased or insufficient for the claim. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accountable decision owner, governance secretary, data owners, records management, Estates information and relevant professional advisers. Required records: decision brief, evidence register, source catalogue, permissions, validation records, current advice and unresolved issue log. Source precedence: current authoritative records and competent advice outrank analytic convenience; no source gains authority merely by being newer. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: owner-defined freshness applies per source and claim; the decision cut-off must not be silently moved. Unresolved conflicts: owners, versions, effective dates, definitions, permissions and professional interpretations; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: claim-specific evidence sufficiency and value-of-information analysis. Decision interval: at each material decision gateway. Minimum evidence: defined decision, options, authority, claim list, decision deadline, consequence of error and an evidence owner for every material claim. Method: classify each gap as missing, stale, conflicting, inaccessible or non-decisive; estimate whether a targeted check could change the option ranking or risk posture. Quantified uncertainty: state residual uncertainty, time-to-evidence and the chance that further evidence changes the decision; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable decision owner and source owner best able to resolve the material gap.

ANSWER BOUNDARY

The service must not infer that more data is always valuable or that an unresolved conflict can be averaged away. It must not disclose privileged advice, personal information or restricted vulnerabilities beyond authorised roles. It must not promise or certify decision readiness, legality, safety or correctness where a material gate remains open; it must not authorise or control delay, approval, escalation, evidence collection or acceptance of residual risk. The smallest adequate remedy may be an owner confirmation, current inspection, limited trial or explicit acknowledgement of uncertainty.

WHY THIS IS USEFUL

The decision owner can commission the smallest justified data or records action instead of delaying indefinitely or collecting unnecessary data.

LB-011

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Can the building absorb the next confirmed peak in teaching, work and visitor demand while preserving current access, service and safety margins?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: overhead thermal entrance counts, anonymous doorway counts, room occupancy state, lift trips, queue-zone counts and key-service load telemetry. Spatial coverage: principal entrances, circulation interfaces, scheduled rooms and shared services in the current capacity model. Observation and aggregation: 1-5 minute peak observations, aggregated into 15 minute demand profiles and representative-day envelopes. Decision-use gate: clock alignment, calibrated counters, at least six peak weeks and 90% valid data during all cited peak windows. Known limits: pass-through traffic, informal demand, simultaneous arrivals and sensor saturation can bias peaks. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: timetabling, School operations, HR workforce planning, events, security, accessibility, fire safety and Estates service owners. Required records: approved timetable, aggregate workforce plans, event schedules, current room capacities, access controls, service limits and competent safety assessments. Source precedence: current legal and competent capacity or egress controls outrank analytic estimates; current schedules outrank the May 2021 floor plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: capacity and route records are stale after layout, closure, works, derating or change of use. Unresolved conflicts: scheduled demand, observed counts, usable capacity, event assumptions and safety margins; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: peak-demand envelope and bottleneck analysis. Decision interval: before each major timetable cycle and large event. Minimum evidence: current capacities, six representative peak weeks, verified route and service mapping, event assumptions and competent constraints. Method: combine scheduled and observed demand, model simultaneous peaks, test plausible surges and identify where usable capacity or service level is exceeded. Quantified uncertainty: show percentile peaks, counter error, scenario range and locations where safety margin cannot be analytically established; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the timetable owner, building operations lead and competent safety or access owner.

ANSWER BOUNDARY

The service must not infer individual attendance, future compliance or safe occupancy from anonymous counts. It must not disclose raw access logs, identifiable event participation or small-group schedules. It must not promise or certify statutory capacity, egress safety or guaranteed service performance; it must not authorise or control admission, timetable, event approval, room capacity or access control. Competent current assessment remains decisive for safety and capacity controls.

WHY THIS IS USEFUL

Leadership can set peak-capacity controls, contingency space and trigger points before confirmed high-demand periods.

LB-012

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

At which entrances, circulation points and lift interfaces do recurring queues or delays materially restrict access, and when do they occur?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: overhead thermal bidirectional counters, non-imaging zone-count sensors, lift-controller status and call/wait aggregates, and door-open duration. Spatial coverage: principal circulation interfaces and accessible routes after current route mapping; exclude raw badge or device traces. Observation and aggregation: 1 minute peak samples, 15 minute summaries and day-type profiles. Decision-use gate: counter calibration at each interface, clocks aligned within 30 seconds and at least 20 comparable peak periods. Known limits: counts cannot measure perceived crowding, mobility needs, cause of delay or uninstrumented route choice. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates operations, accessibility, security, lift maintenance, School operations, timetabling and events owners. Required records: current route and access maps, lift service logs, planned works, timetable, event plans, complaints and competent access assessments. Source precedence: current closure and accessibility instructions outrank telemetry; current route maps outrank the May 2021 plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: route status is stale immediately after closure, works, barrier placement or lift condition change. Unresolved conflicts: sensor counts, lift records, reported experience and different owner definitions of acceptable delay; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: spatiotemporal congestion and queue analysis. Decision interval: monthly, with peak-event assessment. Minimum evidence: current route map, 20 comparable peak periods, valid lift status and an agreed service threshold or experience question. Method: estimate arrival, service and clearance rates; identify repeated pinch points; compare staggering, wayfinding and service scenarios without modelling individuals. Quantified uncertainty: report percentile waits, count error, blind routes and confidence that a pattern is recurrent rather than exceptional; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the building operations lead, accessibility owner and lift or route service owner.

ANSWER BOUNDARY

The service must not infer identity, disability, destination, journey purpose or behaviour from movement counts. It must not disclose raw access events, individual routes or small-group timings. It must not promise or certify safe egress, accessibility compliance or a guaranteed maximum wait; it must not authorise or control route direction, lift priority, timetable, access restriction or works. Emergency and competent access instructions always take precedence.

WHY THIS IS USEFUL

Estates and access owners can target the exact interface, timing and cause requiring a route, lift or operating change.

LB-013

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Does the current portfolio provide enough suitable bookable and informal settings for teaching, collaboration, focus, confidentiality and accessible use?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: booking-panel state, ceiling occupancy state, anonymous doorway counts, room availability and non-content equipment status. Spatial coverage: current room types and informal zones, grouped to suppress small or sensitive use. Observation and aggregation: 5-15 minute samples, daily demand curves and term-level availability by setting type. Decision-use gate: current room classification, four representative weeks per term and at least 85% valid coverage in each reported type. Known limits: occupancy does not reveal activity, suitability, unmet demand, confidentiality or reasons for non-use. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School space lead, central space management, timetabling, room booking, workplace, research facilities and accessibility owners. Required records: current room inventory, capacities, approved functions, booking rules, timetable, equipment lists, access features and aggregate experience evidence. Source precedence: current approved function and owner-confirmed suitability outrank inferred use or historical floor-plan labels. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: room type and feature data are stale after furniture, equipment, access or use changes. Unresolved conflicts: booking categories, observed presence, local ownership, functional suitability and unrecorded informal arrangements; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: setting-type demand, availability and mismatch analysis. Decision interval: termly and before space-planning decisions. Minimum evidence: current taxonomy, four representative weeks, verified capacity and at least one non-sensor source on suitability or unmet need. Method: compare booked, available and anonymously occupied periods by setting type; model reassignment or sharing scenarios while preserving specialist constraints. Quantified uncertainty: show missing demand, classification error, confidence bands and setting types where occupancy is an invalid proxy; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the School space lead and central space-management owner.

ANSWER BOUNDARY

The service must not infer the legitimacy, quality or productivity of work from room use. It must not disclose who booked or occupied a setting, research content or confidential use patterns. It must not promise or certify that low use means surplus or high use means suitable capacity; it must not authorise or control room designation, allocation, booking policy or access. Direct stakeholder evidence and competent suitability assessment remain necessary.

WHY THIS IS USEFUL

Space management can rebalance room types, booking rules or informal provision without interpreting occupancy as activity quality.

LB-014

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Would changing opening hours create enough verified service benefit to justify the associated staffing, security, energy and safeguarding obligations?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous entrance counts, room occupancy state, help-point demand, lift and amenity availability, zone energy load and alarm-arming state. Spatial coverage: the proposed extended-hours zone and its access, welfare and support services. Observation and aggregation: 5-15 minute data by hour and day type, with separate vacation, assessment and event periods. Decision-use gate: at least one comparable trial or 12 months of demand context, 90% valid after-hours coverage and all exceptional closures flagged. Known limits: unobserved unmet demand and staffing or safeguarding quality cannot be derived from counts. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accountable service owner, security, HR, finance, accessibility, safeguarding, sustainability and School operations. Required records: opening policy, staffing models, risk assessments, incident records, transport context, service standards, cost and energy records. Source precedence: safety, labour, equality and safeguarding controls precede demand or energy optimisation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: cost, risk and demand assumptions are stale after service, transport, workforce or programme changes. Unresolved conflicts: observed use, requested access, incremental cost, staff capacity, energy impact and risk controls; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: bounded trial design and net-service-value appraisal. Decision interval: before any permanent change and after a defined trial. Minimum evidence: named owner, control requirements, incremental cost range, demand baseline, trial window, success and stop criteria. Method: compare current and candidate hours, estimate marginal demand and cost, run sensitivity tests and use a reversible trial where evidence is uncertain. Quantified uncertainty: show demand intervals, hidden demand, trial bias and the range of cost or risk assumptions that changes the conclusion; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the service owner, security lead, HR adviser and budget holder.

ANSWER BOUNDARY

The service must not infer purpose of attendance, staff willingness or safe lone working from aggregate presence. It must not disclose individual entry records, staff schedules or small-group routines. It must not promise or certify safety, affordability, equality or success of extended hours; it must not authorise or control opening, staffing, security, access permissions or service levels. Any trial needs current risk controls and an accountable stop decision.

WHY THIS IS USEFUL

The accountable owner can approve a bounded hours trial or retain current hours using a clear net-benefit and duty assessment.

LB-015

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

When a lift, powered door or principal route is unavailable, how much usable capacity is actually lost and which alternative remains workable?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: lift-controller status, powered-door fault and position signals, route-closure status and anonymous aggregate entrance or zone counts. Spatial coverage: each currently mapped accessible journey from entrance to priority destinations, including toilets and refuge arrangements. Observation and aggregation: event-driven status with 1 minute heartbeat, daily availability and incident-duration summaries. Decision-use gate: verified asset-route dependency map, event delivery above 95% and a recent physical validation of each proposed alternative. Known limits: telemetry does not prove route usability for every person, assistance availability or acceptable effort and delay. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accessibility, Estates operations, lift and door maintainers, security, fire safety, School operations and affected service owners. Required records: current access map, outage and maintenance records, assistance arrangements, access assessments, closures and official communications. Source precedence: current competent access advice and official closure status outrank route analytics and the May 2021 floor plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: alternative-route evidence is stale after any physical, operational, staffing or communication change. Unresolved conflicts: asset telemetry, maintainer diagnosis, access assessment and user-reported usability; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: accessible-service dependency and outage-impact analysis. Decision interval: per incident and quarterly for recurring assets. Minimum evidence: current journey map, verified outage, physically checked alternative, assistance owner and destination-service priority. Method: identify destinations rendered unavailable or materially harder to reach, quantify outage duration and compare only professionally validated alternatives. Quantified uncertainty: state affected journey classes, evidence age, unknown user needs and confidence in each alternative; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accessibility lead, building operations lead and affected service owner.

ANSWER BOUNDARY

The service must not infer a person's disability, ability, destination or willingness to use an alternative. It must not disclose individual access records, assistance cases or small-group journeys. It must not promise or certify universal accessibility, compliance or equivalence of an alternative route; it must not authorise or control closure, assistance, access permission, corrective work priority or evacuation procedure. Direct communication and competent assessment remain essential.

WHY THIS IS USEFUL

Continuity and access owners can define usable alternatives, assistance arrangements and escalation thresholds for each critical outage.

LB-016

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which shared amenity is repeatedly failing demand, and is the binding constraint quantity, uptime, replenishment, location or inclusive access?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous doorway or zone counts, equipment run or fault state, fill-level or stock sensors where appropriate, water-use and queue aggregates. Spatial coverage: toilets, catering, printing, lockers, showers, waste points or other named amenities after asset and location mapping. Observation and aggregation: events or 1-15 minute samples, aggregated by hour, day and teaching period. Decision-use gate: verified counter and asset mapping, at least four representative weeks and clear distinction between unavailable, occupied and empty states. Known limits: counts miss perceived quality, cleanliness, hidden demand, non-use and reasons for choosing another amenity. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: amenity service owner, cleaning or catering contractor, Estates operations, accessibility, School operations and procurement. Required records: service standards, asset lists, replenishment schedules, work orders, complaints, inspections and contract performance records. Source precedence: current service standards, competent accessibility evidence and verified asset status outrank inferred dissatisfaction. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: stock, opening, asset and location records are stale after service or layout change. Unresolved conflicts: telemetry, contractor records, inspection findings, complaints and ownership of shared-service failures; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: capacity-availability-demand decomposition. Decision interval: monthly and at contract assessment. Minimum evidence: four representative weeks, current service standard, asset uptime, demand profile and at least one direct quality or experience source. Method: separate queueing, downtime, stock-out, accessibility and locational mismatch; compare service-window and capacity remedies. Quantified uncertainty: report peak ranges, missing demand, sensor ambiguity and confidence that each constraint explains the observed shortfall; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the amenity owner and relevant contractor or Estates service manager.

ANSWER BOUNDARY

The service must not infer satisfaction, personal need, health, gender or protected characteristic from amenity use. It must not disclose individual use, small-group demand or complaint details. It must not promise or certify hygiene, accessibility, contractual compliance or adequate provision; it must not authorise or control service levels, contractor action, stocking, refurbishment or funding. Competent inspection remains necessary for quality and compliance.

WHY THIS IS USEFUL

The service owner can add capacity, retime replenishment, restore uptime or relocate provision according to the verified binding constraint.

LB-017

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Where and when does the aggregate building experience fall below the agreed service level, and what should be investigated before leadership intervenes?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: representative temperature, CO₂, humidity, light and non-recording sound-level telemetry; service availability; anonymous crowding and optional issue-category feedback. Spatial coverage: currently mapped zones with safe aggregation, excluding individual offices and small confidential groups unless combined. Observation and aggregation: 1-5 minute environmental data and daily feedback counts, summarised weekly and by day type. Decision-use gate: calibration state, 85% valid environmental coverage, small-count suppression and at least one direct stakeholder evidence source. Known limits: environmental or crowding signals cannot explain preference, health, satisfaction or the cause of a reported experience. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: student and staff experience leads, School operations, Estates, accessibility, occupational health and service owners. Required records: approved comfort and service standards, aggregate surveys, complaints, work orders, current room functions and competent assessments. Source precedence: direct reported experience and professional advice outrank sensor-only interpretation; standards apply only within their stated scope. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: experience evidence is stale after material service, layout, season or user-population change. Unresolved conflicts: sensor conditions, stakeholder reports, service records and different thresholds for concern; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: time-place experience hotspot detection and evidence triage. Decision interval: monthly and before service-assessment meetings. Minimum evidence: safe aggregation, valid sensor coverage, current zone function, direct experience evidence and a named owner for each issue type. Method: identify repeated co-occurrence, compare with standards and service events, and rank the next investigation by consequence and evidence gap. Quantified uncertainty: show response bias, measurement error, multiple-comparison risk and whether a hotspot is repeated or isolated; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the experience owner and relevant Estates or School service owner.

ANSWER BOUNDARY

The service must not infer individual wellbeing, health, satisfaction, disability or performance from conditions or presence. It must not disclose free-text feedback, personal cases, small groups or identifiable room-use patterns. It must not promise or certify cause, universal experience, regulatory breach or suitability of a remedy; it must not authorise or control service changes, complaints handling, adjustments, staffing or investment. The result should trigger proportionate investigation, not label people or spaces.

WHY THIS IS USEFUL

Leadership can commission the right local investigation rather than acting on an undifferentiated satisfaction score.

LB-018

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What additional space, access and support demand follows from approved changes in student numbers, staffing, programmes, partnerships or hybrid operation?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current aggregate entrance and room demand, occupancy state, service loads, lift trips, amenity use and energy or ventilation capacity indicators. Spatial coverage: the spaces and services affected by each approved change, using current mappings and functional capacities. Observation and aggregation: historic 5-15 minute profiles, term summaries and scenario demand at the planning horizon. Decision-use gate: representative baseline across at least two comparable terms, documented growth assumptions and capacity values validated by owners. Known limits: past demand may not transfer to new pedagogies, schedules, hybrid work, user mix or service design. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: University planning, School executive, admissions, HR, timetabling, space management, research facilities, finance and Estates owners. Required records: approved plans, aggregate headcount and programme assumptions, timetable model, service capacities, access and safety constraints, and project dependencies. Source precedence: approved planning assumptions and competent current capacity constraints outrank extrapolation from historical counts. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: forecasts are stale after plan, programme, workforce, timetable or space changes. Unresolved conflicts: central and School forecasts, headcount definitions, participation assumptions and usable service capacity; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: scenario-based demand and capacity forecasting. Decision interval: at annual planning and each approved-change gateway. Minimum evidence: approved scenario inputs, two comparable terms, current service capacities, uncertainty ranges and owner sign-off for critical assumptions. Method: translate each scenario into demand envelopes, apply scheduling and participation ranges, test bottlenecks and identify trigger points rather than single forecasts. Quantified uncertainty: show low, central and high ranges, model error, non-transferable historical periods and assumptions that dominate the result; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the planning owner, School operations lead and relevant service-capacity owners.

ANSWER BOUNDARY

The service must not infer individual attendance, staff behaviour or programme quality from aggregate demand. It must not disclose personal HR or student data, confidential programme plans or small cohorts beyond authorised aggregation. It must not promise or certify forecast certainty, safe capacity, affordability or inevitability of a project; it must not authorise or control recruitment, staffing, timetable, space allocation, project approval or service reduction. Scenarios are conditional planning evidence, not predictions.

WHY THIS IS USEFUL

Planning can set capacity triggers, phase demand, commission options or revise assumptions before approved growth creates service failure.

LB-019

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Can the proposed event or external-access arrangement proceed without unacceptable displacement of teaching, research, work or inclusive access?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous entrance and zone counts, route and lift status, room occupancy state, queue aggregates, service loads and environmental conditions. Spatial coverage: event spaces, arrival routes, shared interfaces and protected operational zones in the current event plan. Observation and aggregation: 1 minute to 15 minute event-window data, with pre-event baseline and post-event clearance period. Decision-use gate: current event-to-space mapping, calibrated counts, explicit maximum assumptions and verified availability of required services. Known limits: actual arrivals, behaviour, weather, transport and concurrent disruption can differ from the plan. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: event sponsor, School operations, security, fire safety, accessibility, timetabling, research facilities, communications and Estates. Required records: approved event plan, risk assessment, capacities, access arrangements, timetable, bookings, incident command route and service constraints. Source precedence: competent safety, security and accessibility decisions outrank analytic convenience; current closures and operations outrank historical maps. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: event evidence is stale after attendance, layout, route, activity, timing or service changes. Unresolved conflicts: event assumptions, room bookings, operational protections, visitor access and competent constraints; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: event-demand, interface and disruption scenario analysis. Decision interval: before approval and during post-event learning. Minimum evidence: current event plan, attendance range, route and service capacities, protected activities, contingency and competent approvals. Method: simulate arrival and dispersal ranges, identify shared-service overlap, test staggering and separation, and state residual interfaces for owner judgement. Quantified uncertainty: show attendance range, peak sensitivity, unmodelled behaviour and conditions under which the plan no longer applies; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the event sponsor and each competent safety, access and operational owner.

ANSWER BOUNDARY

The service must not infer participant identity, purpose, behaviour or compliance from aggregate counts. It must not disclose guest lists, access credentials, protected research details or small-group movements. It must not promise or certify event safety, security, accessibility or absence of disruption; it must not authorise or control event approval, admission, route control, timetable, staffing or incident action. On-site command and official instructions take precedence during the event.

WHY THIS IS USEFUL

The event owner can proceed, modify, relocate or decline the arrangement with displacement, access and control duties explicit.

LB-020

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which access and experience actions should enter the next funded programme, and how stable is their priority under different stakeholder weightings?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: aggregate barrier reports, lift and door availability, route status, amenity demand, environmental conditions, queue patterns and repeat fault telemetry. Spatial coverage: only candidate locations with current ownership, mapping and a defined service or experience outcome. Observation and aggregation: monthly patterns and multi-term recurrence, with incident-level evidence for high-consequence failures. Decision-use gate: at least two evidence types, a verified current condition, safe aggregation and a named benefit and maintenance owner. Known limits: reported need, hidden demand, unequal feedback access and low-frequency severe barriers may not be captured proportionately. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accessibility and inclusion, student and staff experience, Estates, School operations, finance, risk, sustainability and affected service owners. Required records: current access assessments, complaints, service standards, work orders, condition surveys, costs, dependencies and approved strategies. Source precedence: mandatory safety and access duties and competent assessments precede discretionary ranking. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: condition and cost evidence is stale after deterioration, corrective work, scope or market change. Unresolved conflicts: severity, recurrence, cost, affected service, lived experience, deliverability and owner priorities; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: constrained portfolio prioritisation. Decision interval: annually and at programme refresh. Minimum evidence: defined candidate set, current condition, competent duty check, cost range, affected-service evidence, dependencies and benefit owner. Method: screen mandatory work first, then compare remaining items across consequence, recurrence, inclusion, service value, cost and deliverability; test alternative weightings. Quantified uncertainty: show ranges, missing voices, sensitivity, unquantified dignity or social value and projects whose order is unstable; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable programme owner and relevant competent or service owners.

ANSWER BOUNDARY

The service must not infer the value of a person or group, prevalence of need or fairness from complaint volume. It must not disclose case details, protected characteristics, individual adjustments or security-sensitive defects. It must not promise or certify equality compliance, value for money, deliverability or the correct investment order; it must not authorise or control programme, funding, procurement, scope or risk acceptance. Low-volume high-consequence barriers must remain visible.

WHY THIS IS USEFUL

The programme sponsor can form a transparent, fundable sequence of actions and show which priorities are sensitive to weighting.

3 Operations, teaching, research and workplace performance

Stakeholder 32 | LB-021 - LB-022

LB-021

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are the rooms scheduled for the next teaching period genuinely ready for access, environment, furniture, power and teaching technology before first use?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: door and powered-door status, room occupancy state before session, representative temperature and CO2, lighting and power state, teaching-technology heartbeat and BMS alarms. Spatial coverage: scheduled teaching rooms and their accessible approach, excluding cameras, microphones and content inspection. Observation and aggregation: 1-5 minute checks from 30 minutes before to 15 minutes after scheduled start, aggregated by room and teaching week. Decision-use gate: current timetable and room mapping, commissioned endpoints, clock alignment within one minute and valid status for every claimed readiness component. Known limits: furniture arrangement, cleanliness, specialist setup and subtle usability faults may need physical inspection. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: timetabling, School teaching operations, audiovisual support, Estates, accessibility, cleaning and room owners. Required records: current timetable, room specification, teaching-technology inventory, service standards, fault tickets, planned works and accessibility information. Source precedence: current owner-confirmed room status and official relocation instructions outrank telemetry; the May 2021 floor plan cannot establish readiness. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: room and equipment data is stale after schedule, layout, project, equipment or closure change. Unresolved conflicts: timetable, booking, AV inventory, BMS status, cleaning records and local room checks; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: pre-session readiness validation and exception routing. Decision interval: before each scheduled session, with weekly trend assessment. Minimum evidence: current session-room link, valid status for all critical components, known support owner and a rule for components requiring human inspection. Method: apply a room-specific checklist, distinguish verified ready, verified not ready and unknown, and prioritise by time to session and available alternative. Quantified uncertainty: show component-level freshness, missing checks, false-alarm history and confidence that the issue affects the scheduled use; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the teaching operations lead and relevant AV, Estates, cleaning or accessibility owner.

ANSWER BOUNDARY

The service must not infer teaching quality, staff preparation, attendance or learning from room conditions. It must not disclose class lists, teaching content, individual presence or restricted session details. It must not promise or certify fitness for every learner, complete readiness or absence of hidden defects; it must not authorise or control room release, timetable change, support dispatch, access or equipment control. A competent or local check remains necessary where telemetry cannot verify the requirement.

WHY THIS IS USEFUL

Teaching and Estates teams can clear readiness exceptions before use, provide alternatives and focus investment on recurrent preparation failures.

LB-022

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which recurring building or room conditions are responsible for the largest verified loss of teaching time, late starts or forced relocations?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: room environmental and power telemetry, AV heartbeat, door and lift status, service alarms and anonymous occupancy-state transitions. Spatial coverage: scheduled teaching spaces and essential routes with current service mapping; no lesson content or person-level attendance. Observation and aggregation: event data aligned to session windows, weekly summaries and term-level recurrence profiles. Decision-use gate: at least 10 comparable incidents or two terms of data, clock alignment, verified incident labels and 90% valid telemetry around cited events. Known limits: late starts may arise from pedagogy, staffing, timetable or other causes not visible to building systems. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School teaching operations, timetabling, AV support, Estates service desk, accessibility and room owners. Required records: verified disruption logs, room moves, fault tickets, timetable, planned works, service restoration and local confirmation. Source precedence: verified disruption records and owner-confirmed incident cause outrank inferred association from telemetry. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: incident labels are stale when later investigation changes cause, scope or duration. Unresolved conflicts: teaching reports, service tickets, BMS or AV alarms, timetable changes and causal classification; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: incident linkage, recurrence and contributing-condition analysis. Decision interval: monthly and at term assessment. Minimum evidence: verified disruption definition, aligned timestamps, at least 10 events where rate estimation is attempted, comparator sessions and current room mapping. Method: link service events to session disruption, calculate recurrence and duration, compare with unaffected sessions, and keep confirmed causes separate from associations. Quantified uncertainty: report event counts, rate intervals, unmatched events, confounding timetable effects and confidence in causal classification; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the teaching operations owner and relevant service manager.

ANSWER BOUNDARY

The service must not infer staff performance, teaching quality, student behaviour or responsibility from delay patterns. It must not disclose individual session participants, teaching content or identifiable staff patterns. It must not promise or certify causation unless confirmed by the accountable investigation, or that all disruption is captured; it must not authorise or control room moves, maintenance priority, compensation, staffing or service changes. Small event counts should be described, not converted into unstable rankings.

WHY THIS IS USEFUL

Leadership can target the failure mode with the largest verified teaching consequence and test whether corrective action reduces recurrence.

3 Operations, teaching, research and workplace performance

Stakeholder 32 | LB-023 - LB-024

LB-023

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which research-support spaces, utilities or technical services are below the reliability and recovery level required by approved research dependencies?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: asset run-state, power quality, temperature and humidity, differential pressure where required, gas or water service status, UPS alarms and non-content equipment heartbeat. Spatial coverage: only shared research-support assets and rooms with owner-approved telemetry; exclude research data, samples, images, code and instrument output content. Observation and aggregation: seconds to 5 minute sampling according to service criticality, with daily availability and incident-duration summaries. Decision-use gate: current asset-to-service dependency map, calibrated environmental points, alarm-path tests and owner-defined uptime thresholds. Known limits: telemetry cannot establish experiment validity, sample integrity or suitability for every protocol. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: research facilities managers, principal service owners, School research leadership, Estates, safety, IT and equipment custodians. Required records: current asset register, service-level requirements, maintenance and calibration records, approved operating procedures, outage logs and recovery plans. Source precedence: equipment custodian and competent safety or technical requirements outrank generic building thresholds. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: requirements and dependencies are stale after protocol, equipment, service, layout or ownership change. Unresolved conflicts: facility records, Estates services, equipment logs, local operating requirements and responsibility for recovery; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: service availability, stability and dependency analysis. Decision interval: weekly for critical services and quarterly for portfolio assessment. Minimum evidence: approved service requirement, current dependency map, validated telemetry, incident definitions and owner-confirmed recovery objective. Method: calculate availability and interruption duration, detect repeated instability, map upstream dependencies and test recovery against the stated service objective. Quantified uncertainty: show coverage, event-detection limits, rate intervals and services where research consequence cannot be quantified safely; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the research facilities owner and relevant Estates, IT or technical service owner.

ANSWER BOUNDARY

The service must not infer research quality, productivity, protocol compliance or sample integrity from building telemetry. It must not disclose research content, project identity, instrument output or small-team activity. It must not promise or certify fitness for research, regulatory compliance or protection of samples and experiments; it must not authorise or control research activity, equipment, utilities, maintenance or recovery action. The research owner must interpret consequence and approve protective action.

WHY THIS IS USEFUL

Research and Estates owners can protect critical dependencies, set recovery requirements and prioritize reliability work without inspecting research content.

LB-024

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Does the workplace supply the agreed mix of focused, collaborative, confidential and accessible settings without treating presence as productivity?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: ceiling occupancy-state sensors, anonymous doorway counts, booking-panel status, non-recording sound-level measures and representative environmental conditions. Spatial coverage: workplace setting types aggregated across safe zones; exclude personal desks, device telemetry and content capture. Observation and aggregation: 5-15 minute samples, daily availability and weekly setting-type profiles. Decision-use gate: current setting taxonomy, four representative weeks, 85% valid coverage and small-zone suppression. Known limits: presence and sound level cannot reveal task, confidentiality, preference, productivity or whether a setting was suitable. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School leadership, workplace and HR policy owners, space management, accessibility, occupational health and Estates. Required records: approved workplace model, current room functions, booking rules, accessibility features, aggregate surveys, complaints and service standards. Source precedence: approved workplace and adjustment arrangements plus direct experience evidence outrank inferred suitability. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: setting classification is stale after layout, furniture, acoustic, access or policy change. Unresolved conflicts: space labels, observed use, booking data, local arrangements and stakeholder experience; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: setting-supply, availability and experience-gap analysis. Decision interval: termly and before workplace changes. Minimum evidence: current setting definitions, representative weeks, safe aggregation, direct experience evidence and named owner for each setting type. Method: compare availability and demand by setting type, identify recurring shortages or unused capacity, and test sharing or conversion scenarios. Quantified uncertainty: show missing demand, preference heterogeneity, classification error and cases where occupancy is not a valid suitability measure; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the workplace owner, School space lead and accessibility adviser.

ANSWER BOUNDARY

The service must not infer individual productivity, effort, attendance, collaboration quality or confidentiality from space use. It must not disclose desk-level activity, personal bookings, device identifiers or small-team routines. It must not promise or certify workplace effectiveness, wellbeing, equality or suitability for an individual adjustment; it must not authorise or control attendance policy, performance management, allocation, booking rules or refurbishment. Direct engagement and individual adjustment processes remain separate.

WHY THIS IS USEFUL

School leadership can rebalance settings, booking rules and support provision while protecting confidential, accessible and specialist needs.

3 Operations, teaching, research and workplace performance

Stakeholder 32 | LB-025 - LB-026

LB-025

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which operational failure modes create the greatest verified mission loss after frequency, duration and consequence are considered together?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: BMS and equipment alarms, power and network availability, lift and door status, environmental excursions and anonymous zone-availability indicators. Spatial coverage: services and locations in the current incident taxonomy, excluding personal or content-level evidence. Observation and aggregation: event-driven telemetry with one-minute clock alignment, incident snapshots and monthly consequence summaries. Decision-use gate: owner-confirmed incident start, end, affected service and restoration, plus at least 95% event-path health for ranked incident types. Known limits: low-frequency severe events, unreported disruption and shared causes can distort simple counts. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates operations, School operations, IT, research facilities, teaching support, security and business-continuity owners. Required records: incident records, service desk tickets, outage communications, restoration logs, timetable or booking impacts and owner-confirmed consequence. Source precedence: formal incident record and accountable service-owner classification outrank automatic alarm correlation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: incident data is stale when post-incident assessment changes duration, cause or consequence. Unresolved conflicts: alarm events, service tickets, operational impact, root-cause status and cross-service ownership; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: incident consequence, recurrence and Pareto analysis. Decision interval: monthly and annually. Minimum evidence: common incident taxonomy, verified service-loss minutes, affected-service owner, severity rule and sufficient events for any rate comparison. Method: sum verified service consequence, separate frequency from severity, group only owner-confirmed common causes and identify repeat restorations without durable closure. Quantified uncertainty: show event counts, duration ranges, under-reporting risk, attribution class and uncertainty for rare events; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the building operations lead and accountable owner of each high-consequence service.

ANSWER BOUNDARY

The service must not infer individual responsibility, staff performance or research loss value from incident timing. It must not disclose personal case data, sensitive vulnerabilities or restricted research consequence. It must not promise or certify root cause, complete incident capture, future reliability or adequacy of controls; it must not authorise or control maintenance, staffing, compensation, risk acceptance or service restoration. High consequence and low frequency must remain visible alongside common faults.

WHY THIS IS USEFUL

Operations leaders can rank failure modes by mission consequence, assign root-cause work and avoid focusing only on high ticket counts.

LB-026

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are cleaning, waste, catering, reception and other support services deployed where and when demand and contractual service levels require them?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous zone counts, bin fill-level sensors, washroom supply state, service-point queue counts, equipment uptime and environmental or hygiene proxy telemetry where approved. Spatial coverage: named public and shared zones only, with no imaging, audio or individual service-use tracking. Observation and aggregation: 5-30 minute operational data and daily or weekly service-level summaries. Decision-use gate: current zone and contract mapping, at least four representative weeks, verified sensor limitations and human inspection for quality claims. Known limits: sensors cannot certify cleanliness, food safety, customer service quality or hidden demand. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates soft-services, procurement, contractors, School operations, catering, waste, reception, accessibility and health-and-safety owners. Required records: current service specification, schedules, inspections, work orders, complaint categories, replenishment and contractor performance records. Source precedence: competent inspection and approved contractual measures outrank telemetry-only service judgements. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: service schedules and standards are stale after contract, opening-hour, occupancy-pattern or zone-use change. Unresolved conflicts: contractor returns, inspections, telemetry, complaints and local reports of service quality; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: demand-service alignment and service-level exception analysis. Decision interval: weekly operationally and monthly at contract assessment. Minimum evidence: current standard, four representative weeks, verified demand proxy, completed inspections and a named service owner. Method: compare demand profile with scheduled delivery, separate capacity, timing, uptime and quality failures, and test revised service windows before expanding resource. Quantified uncertainty: show demand ranges, inspection sampling limits, missing zones and confidence that timing rather than quality explains the gap; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the soft-service owner and relevant contractor manager.

ANSWER BOUNDARY

The service must not infer individual behaviour, hygiene, satisfaction or responsibility from service demand. It must not disclose personal complaints, identifiable use or small-group patterns. It must not promise or certify contractual compliance, cleanliness, food safety or adequate staffing; it must not authorise or control contract action, staffing, service schedule, procurement or payment. Human inspection remains authoritative for quality and safety.

WHY THIS IS USEFUL

The contract or service owner can retime, resize or challenge delivery against demand and agreed outcomes rather than anecdote.

LB-027

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

When is the least disruptive window for planned maintenance, testing or project work, and what contingency is required if the work overruns?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: aggregate room and entrance demand, service-load profiles, asset run-state, alarm history, route availability and environmental conditions. Spatial coverage: affected assets, spaces, routes and downstream services in the verified work dependency map. Observation and aggregation: historic 5-15 minute profiles, scheduled-work windows and event snapshots. Decision-use gate: current dependency mapping, representative demand for the proposed period and verified telemetry health for each critical service. Known limits: unexpected work duration, concealed defects, academic changes and user-specific needs remain uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: maintenance planner, project manager, School operations, timetabling, research facilities, IT, security, accessibility and safety owners. Required records: approved work scope, method statement, permits, asset dependencies, timetable, bookings, research constraints, communications and contingency plans. Source precedence: competent safety controls, permits and current service-owner constraints outrank low-demand analytics. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: work and dependency evidence is stale after scope, method, schedule, occupancy, route or asset-status change. Unresolved conflicts: contractor plan, service-owner constraints, timetable, research protections and contingency capability; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: maintenance-window optimisation with constraint and contingency testing. Decision interval: for each material work package. Minimum evidence: current work scope, dependencies, competent controls, demand envelope, protected activities, outage estimate and tested contingency owner. Method: screen infeasible windows, estimate mission exposure for remaining windows, test overrun scenarios and present the trade-off rather than a single automatic date. Quantified uncertainty: show duration range, demand uncertainty, unmodelled specialist constraints and the share of tested cases in which contingency capacity is exceeded; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the maintenance or project owner and each affected service owner.

ANSWER BOUNDARY

The service must not infer that low occupancy means work is safe or consequence-free. It must not disclose confidential schedules, research content, personal adjustments or restricted security details. It must not promise or certify work safety, permit validity, completion time or adequacy of contingency; it must not authorise or control work approval, outage, contractor instruction, timetable or service suspension. Competent persons and affected owners retain stop-work and approval authority.

WHY THIS IS USEFUL

The work sponsor can choose a low-disruption window, publish contingencies and protect critical activities if duration or scope changes.

LB-028

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which faults are detected late, wait longest for accountable action, or recur after closure without a durable resolution?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: BMS alarms, asset run-state, environmental excursions, vibration or temperature condition points where justified, help-desk timestamps and restoration status. Spatial coverage: mapped critical assets and downstream services, with validated alarm routes and no personal performance telemetry. Observation and aggregation: seconds to 5 minute events, full incident timelines and monthly recurrence summaries. Decision-use gate: known clock offsets, alarm-path tests, owner-confirmed detect, acknowledge, attend, restore and close times. Known limits: silent failures, alarm suppression, manual work and ambiguous closure codes may hide delay or recurrence. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates operations, maintenance contractors, service desk, asset owners, School operations and affected service owners. Required records: asset register, criticality, alarm configuration, work orders, attendance records, restoration tests, root-cause and repeat-fault history. Source precedence: verified work-order and asset-owner status outrank an alarm-only inference; final cause remains with competent investigation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: fault status is stale when later testing reopens or reclassifies the issue. Unresolved conflicts: alarm timestamps, service-desk handling, contractor records, root cause and definition of durable closure; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: fault lifecycle, delay decomposition and repeat-failure analysis. Decision interval: weekly for critical faults and monthly for portfolio assessment. Minimum evidence: current asset mapping, complete timestamps, agreed lifecycle states, verified restoration and sufficient repeat events for recurrence claims. Method: calculate delay by lifecycle stage, identify recurrent assets and distinguish temporary restoration, duplicate tickets and confirmed durable corrective work. Quantified uncertainty: show missing timestamps, censoring, small-event uncertainty and confidence in root-cause grouping; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the maintenance service owner and accountable asset owner.

ANSWER BOUNDARY

The service must not infer individual staff or contractor performance, negligence or intent from response times. It must not disclose personal work records, security-sensitive alarm paths or commercial details beyond authorised roles. It must not promise or certify root cause, contractual breach, competence or future reliability; it must not authorise or control disciplinary action, contract remedy, dispatch, maintenance priority or asset shutdown. Process evidence should prompt owner assessment, not personnel surveillance.

WHY THIS IS USEFUL

Estates leadership can remove detection and triage delays, reopen unresolved cases and commission durable fault elimination.

LB-029

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which dependencies among power, network, controls, ventilation, access and specialist services could turn a local fault into a wider outage?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: time-synchronised power, UPS, network, BMS, ventilation, lift, door and critical-equipment availability events. Spatial coverage: only owner-approved service interfaces in a verified dependency graph; restrict detailed topology to authorised roles. Observation and aggregation: seconds to one-minute events with incident snapshots and quarterly cascade summaries. Decision-use gate: clock alignment within the causal resolution required, tested signal delivery and owner-validated dependency edges. Known limits: coincident alarms do not prove direction, common cause or complete topology. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates engineering, IT, security, research facilities, maintenance and business-continuity owners. Required records: current service topology, asset dependencies, incident investigations, configuration and change records, maintenance tests and restoration sequences. Source precedence: current technical owner diagrams and confirmed incident findings outrank statistical ordering or historical floor-plan assumptions. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: dependency data is stale after configuration, project, bypass, temporary supply or operating-mode change. Unresolved conflicts: technical diagrams, event order, common-cause findings, ownership boundaries and restricted vulnerability details; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: dependency-graph incident reconstruction and common-cause screening. Decision interval: after each material incident and quarterly. Minimum evidence: validated topology, aligned events, owner-confirmed incident boundary and at least two corroborating sources for a claimed cascade edge. Method: reconstruct event order, test known dependencies and common causes, assign evidence strength to each edge and identify single points or circular recovery dependencies. Quantified uncertainty: show timing tolerance, unobserved components, competing causal graphs and confidence by dependency edge; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the relevant engineering or IT service owner and business-continuity lead.

ANSWER BOUNDARY

The service must not infer causation from temporal order alone or responsibility across service teams. It must not disclose restricted network, security or utility topology beyond need-to-know roles. It must not promise or certify complete dependency coverage, root cause or resilience of the service chain; it must not authorise or control isolation, configuration, maintenance, restoration order or investment. Competent technical investigation remains authoritative.

WHY THIS IS USEFUL

Owners can identify cascade paths, strengthen isolation and monitoring, and act at the dependency with the greatest systemic effect.

LB-030

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Did the recent operational change produce a sustained service benefit after timetable, weather, season and concurrent changes are accounted for?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: the change-specific aggregate service outcome, demand context, environmental, fault, energy and availability telemetry. Spatial coverage: intervention and suitable comparison assets or zones with stable mappings. Observation and aggregation: matched pre- and post-change periods, daily or weekly summaries and follow-up long enough to test persistence. Decision-use gate: documented change date, at least eight comparable weeks before and after where feasible, 90% valid outcome coverage and no unrecorded concurrent change. Known limits: unmeasured operational changes, learning effects and changing demand can bias attribution. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: change owner, affected service owner, School operations, Estates, finance, sustainability, accessibility and change-control owners. Required records: approved change record, objective, baseline, implementation evidence, concurrent changes, service tickets, costs and stakeholder feedback. Source precedence: verified implementation and owner-defined outcome precede analytic interpretation; competent safety or access evidence constrains success claims. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: baseline is stale if scope, outcome definition or operating context changes materially. Unresolved conflicts: implementation completeness, concurrent change, outcome definition, cost and stakeholder experience; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: interrupted time-series or matched-comparison evaluation. Decision interval: after the agreed stabilisation period and at benefits assessment. Minimum evidence: defined intervention date, stable outcome, adequate pre/post observations, demand and season covariates, and a credible comparison or counterfactual rationale. Method: estimate level and trend change, test persistence, inspect pre-trends and conduct sensitivity analysis excluding abnormal periods. Quantified uncertainty: report effect interval, model assumptions, residual confounding and whether evidence supports association or a stronger contribution claim; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the change owner and accountable service or benefit owner.

ANSWER BOUNDARY

The service must not infer causation, staff performance or stakeholder benefit from a before-after difference alone. It must not disclose individual behaviour, personal feedback or restricted operational details. It must not promise or certify permanent success, value for money, safety or transferability to other settings; it must not authorise or control continuation, rollback, staffing, contract or future rollout. A null result may reflect inadequate evidence rather than no effect.

WHY THIS IS USEFUL

The change owner can retain, adapt or reverse the intervention on sustained net service effect rather than a simple before-and-after comparison.

LB-031

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which safety-critical system has a current impairment, overdue action or evidence gap that requires immediate attention from its accountable owner?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: only approved fire-alarm, suppression, smoke-control, emergency-lighting, lift, refuge-communication and critical-plant status or fault telemetry. Spatial coverage: systems and zones in the current safety asset register; detailed vulnerabilities restricted to authorised roles. Observation and aggregation: event-driven status with tested heartbeat and current snapshot, plus daily exception summaries. Decision-use gate: commissioned signal path, current asset mapping, 100% reconciliation of active critical impairments with the formal register and owner acknowledgement. Known limits: a normal signal does not prove system effectiveness, test validity, physical condition or regulatory compliance. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: responsible person or duty holder, fire and safety managers, Estates engineering, specialist maintainers and building operations. Required records: formal impairment register, statutory and planned test records, certificates, permits, compensatory measures, work orders and competent advice. Source precedence: current formal safety records and competent professional instruction outrank analytics or telemetry. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: critical status is stale at the next state change, test, impairment, isolation or compensatory-measure change. Unresolved conflicts: telemetry, formal registers, maintainer diagnosis, certificate scope and ownership of compensatory action; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: critical-impairment reconciliation and overdue-action exception checking. Decision interval: continuous operational assessment and formal governance cadence. Minimum evidence: current formal register, competent owner, verified system status, action due date and compensatory controls for every cited item. Method: reconcile telemetry and formal records, flag unmatched or overdue states and present consequence class without replacing professional judgement. Quantified uncertainty: state source age, unmonitored components, certificate scope and any unresolved discrepancy as an assurance gap; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the duty holder, safety manager and relevant competent maintainer.

ANSWER BOUNDARY

The service must not infer safety, compliance, system effectiveness or acceptable risk from telemetry status. It must not disclose security-sensitive vulnerabilities, personal evacuation information or restricted system details beyond authorised roles. It must not promise or certify legal compliance, safe occupation or adequacy of compensatory measures; it must not authorise or control isolation, reopening, evacuation, maintenance, risk acceptance or incident action. Current competent instruction and incident command take precedence.

WHY THIS IS USEFUL

The accountable safety owner can verify the impairment, apply current controls, restore assurance and communicate only the supported status.

LB-032

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Do current layouts and aggregate demand indicate that a competent assessment of occupancy control, circulation or egress capacity is now required?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous entrance and zone counts, doorway flow counts, lift status and route-availability events. Spatial coverage: currently mapped spaces and circulation interfaces; no person-level access logs or evacuation tracking. Observation and aggregation: 1 minute peak data, 15 minute summaries and event-specific demand envelopes. Decision-use gate: calibrated counters, current layout and capacity mapping, at least 20 comparable peak periods and known sensor saturation limits. Known limits: normal-operation counts cannot demonstrate emergency movement, behaviour, evacuation time or statutory capacity. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: responsible person, fire safety, Estates, space management, security, accessibility, School operations and event owners. Required records: current fire strategy, competent occupancy and egress assessments, layout, room use, event plans, capacities and official restrictions. Source precedence: competent current fire and access assessments outrank sensor-derived demand; the May 2021 plan has no current assurance status. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: layout and capacity evidence is stale immediately after physical or use change. Unresolved conflicts: observed demand, approved use, competent capacity, temporary arrangements and route availability; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: peak-demand screening against current competent control assumptions. Decision interval: termly and before major events or change of use. Minimum evidence: current layout and use, competent control values, calibrated counts, representative peaks and explicit screening threshold. Method: compare observed demand ranges with the assumptions underlying current controls and flag only cases needing competent assessment. Quantified uncertainty: show count error, unobserved routes, scenario range and the distance from - not compliance with - the competent control value; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the responsible person and competent fire, access or crowd-safety adviser.

ANSWER BOUNDARY

The service must not infer safe occupancy, evacuation performance or individual compliance from normal-operation counts. It must not disclose raw access logs, individual routes or security-sensitive plans beyond authorised roles. It must not promise or certify statutory capacity, egress safety, fire-strategy validity or need for closure; it must not authorise or control occupancy limits, event approval, layout, access or evacuation action. The output is a assessment trigger only.

WHY THIS IS USEFUL

The duty holder can commission a competent capacity or egress assessment without treating sensor counts as certification.

LB-033

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Where are environmental conditions repeatedly outside current approved limits, for how long, and which competent owner must investigate?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: calibrated room temperature, relative humidity, CO2 and, where justified, particulate or differential-pressure sensors; BMS setpoint and plant status. Spatial coverage: representative locations selected for each room type, avoiding direct supply air, heat sources and unrepresentative corners. Observation and aggregation: 1-5 minute samples, hourly distributions and daily excursion summaries. Decision-use gate: documented siting, calibration, clock alignment, 90% valid data and owner-approved thresholds for the stated purpose. Known limits: spot sensors may miss spatial variation; CO2 is not a direct measure of infection risk or full ventilation performance. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates engineering, health and safety, occupational health, research facilities, room owners and competent ventilation or environmental specialists. Required records: current design and operating criteria, risk assessments, maintenance, calibration, complaints, investigations and any activity-specific requirements. Source precedence: competent current requirements and investigation findings outrank generic thresholds or historic room labels. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: limits and room requirements are stale after use, occupancy, ventilation or process change. Unresolved conflicts: sensor readings, BMS values, complaints, maintenance status and activity-specific criteria; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: threshold-excision, duration and contributing-condition analysis. Decision interval: weekly for monitored critical spaces and monthly elsewhere. Minimum evidence: approved threshold, calibrated representative sensor, current room use, sufficient valid coverage and owner-confirmed plant or service context. Method: calculate frequency, duration and severity of excursions, compare with plant state and demand, and identify when physical investigation is needed. Quantified uncertainty: show measurement tolerance, spatial uncertainty, missing periods and confidence that the excursion is real and recurrent; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the relevant Estates engineer, health-and-safety owner or competent specialist.

ANSWER BOUNDARY

The service must not infer individual health, exposure, comfort, compliance or cause from environmental telemetry alone. It must not disclose personal health information, identifiable occupancy or research-process details. It must not promise or certify safe conditions, regulatory compliance, ventilation adequacy or absence of risk; it must not authorise or control plant settings, room closure, work restriction or remedial action. Competent assessment and current activity knowledge remain necessary.

WHY THIS IS USEFUL

The relevant competent owner can investigate repeated exceedance, protect occupants and distinguish plant, load, weather and sensor causes.

LB-034

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are all required inspections, tests, certificates and resulting actions current, valid, correctly scoped and traceable to the right asset?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: asset identity, service-state and test-due telemetry only where the authoritative compliance system exposes an approved status interface. Spatial coverage: assets in the current statutory and policy register with verified identifiers and ownership. Observation and aggregation: event updates on test or status change and daily exception snapshots. Decision-use gate: complete asset reconciliation, owner-confirmed source interface and no substitution of telemetry for the signed record. Known limits: an interface can omit scope, limitations, conditions, invalidation or documents held outside the system. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: duty holders, compliance managers, Estates asset information, competent inspectors, specialist contractors and records management. Required records: current statutory register, signed reports, certificates, test results, asset mapping, action tracker, exemptions and competent advice. Source precedence: signed current records and duty-holder interpretation outrank dashboard status, imported dates or sensor evidence. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: a record is stale or invalid after expiry, material asset change, failed test, superseding advice or scope mismatch. Unresolved conflicts: asset identity, certificate scope, due dates, imported status, action closure and competent interpretation; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: asset-to-record reconciliation, validity and overdue-action checking. Decision interval: daily exception assessment and formal assurance cycles. Minimum evidence: current asset register, signed source record, scope, effective and expiry dates, action owner and evidence of closure. Method: match each obligation to the correct asset, verify scope and validity, distinguish missing, expired, superseded and conditionally valid records. Quantified uncertainty: state unmatched assets, inaccessible records, uncertain identity and any action whose closure evidence is incomplete; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the duty holder, compliance manager and competent inspector.

ANSWER BOUNDARY

The service must not infer compliance, safety or asset condition from a green dashboard field. It must not disclose restricted certificates, personal signatures or security-sensitive details beyond authorised roles. It must not promise or certify legal compliance, validity outside stated scope or adequacy of a corrective action; it must not authorise or control certificate acceptance, asset use, closure, maintenance or risk acceptance. Only the authoritative record and competent owner can establish status.

WHY THIS IS USEFUL

Duty holders can close document-to-asset gaps, complete overdue actions and avoid relying on certificates that are stale or out of scope.

LB-035

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are inclusive evacuation, refuge, communication and assistance arrangements current, usable and matched to the activities planned in the building?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: refuge communication health, evacuation-lift or lift status where applicable, powered-door status, alarm and visual-alert status, and route-closure signals. Spatial coverage: current inclusive evacuation routes and systems; no personal tracking or exposure of individual assistance plans. Observation and aggregation: event-driven status with daily heartbeat and pre-event or pre-occupation checks. Decision-use gate: verified system-to-route mapping, successful communication test and reconciliation with current formal arrangements. Known limits: telemetry cannot prove an arrangement is suitable for a particular person, staffed, understood or executable under incident conditions. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: responsible person, fire safety, accessibility, security, HR or student-support owners, School operations and affected individuals through protected processes. Required records: current evacuation strategy, Personal Emergency Evacuation Plan governance, refuge and communication tests, event plans, training and assistance arrangements. Source precedence: current competent evacuation advice, protected person-owned arrangements and incident command outrank analytics. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: arrangements are stale after person, activity, layout, route, staffing or system change. Unresolved conflicts: system status, formal strategy, event plan, confidential assistance arrangements and operational staffing; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: arrangement completeness and system-readiness exception checking. Decision interval: before major activities and at the responsible person's assessment cadence. Minimum evidence: current strategy, verified systems, route and staffing map, protected confirmation that required arrangements exist, and competent owner sign-off. Method: check that each required function has a current owner and tested route without exposing personal details; list unknowns and failed dependencies. Quantified uncertainty: state evidence age, untested scenarios, unavailable systems and aspects that only an individual or competent adviser can validate; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the responsible person, fire safety and accessibility leads.

ANSWER BOUNDARY

The service must not infer disability, identity, location or assistance need of any person. It must not disclose Personal Emergency Evacuation Plans, protected case data or security-sensitive route details beyond authorised roles. It must not promise or certify safe evacuation, individual suitability, compliance or readiness during a real incident; it must not authorise or control evacuation, assistance, lift use, building access or incident response. Official alarm, on-scene command and current personal arrangements take precedence.

WHY THIS IS USEFUL

Emergency and inclusion owners can update arrangements, test interfaces and ensure current activities have workable assistance and communication.

LB-036

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Do incidents and near misses reveal a recurring hazard pattern by location, activity period or service interface that merits formal investigation?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: approved aggregate environmental, slip-risk, leak, door, lift, alarm, service-state and anonymous zone-demand telemetry relevant to each incident type. Spatial coverage: only locations and periods safely aggregated; no individual monitoring or reconstruction of events from content. Observation and aggregation: event windows aligned to incident times, monthly summaries and multi-year context for rare hazards. Decision-use gate: verified incident labels, current location mapping, sufficient events for rate analysis and documented sensor health around each event. Known limits: near-miss reporting is incomplete and biased; telemetry may miss human, procedural and contextual factors. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: health and safety, responsible person, School operations, Estates, occupational health and relevant service owners. Required records: formal incident and near-miss records, investigations, risk assessments, work orders, training and corrective-action logs. Source precedence: formal investigation and competent risk assessment outrank statistical association. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: incident classification is stale when investigation changes hazard, location, cause or consequence. Unresolved conflicts: reporting completeness, incident taxonomy, sensor context, corrective-action ownership and disputed causal factors; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: pattern detection with exposure-aware rate and severity analysis. Decision interval: quarterly and after any high-consequence event. Minimum evidence: common taxonomy, verified events, defensible exposure denominator where rates are used, current mapping and enough events to avoid unstable small-number ranking. Method: separate frequency, severity and exposure; use control charts or exact intervals; identify clusters for competent investigation, not automatic cause assignment. Quantified uncertainty: show event counts, reporting bias, rate intervals, multiple-testing risk and unmeasured exposure; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the health-and-safety owner and competent investigator for the hazard.

ANSWER BOUNDARY

The service must not infer individual fault, unsafe behaviour, protected characteristics or causation from a statistical cluster. It must not disclose personal incident details, health information or security-sensitive evidence. It must not promise or certify risk level, legal breach, root cause or adequacy of controls; it must not authorise or control disciplinary action, risk acceptance, closure, work restriction or corrective measures. Rare severe events must not be dismissed because a stable rate cannot be estimated.

WHY THIS IS USEFUL

Safety leaders can commission a targeted investigation and corrective programme where recurrence and consequence exceed the agreed trigger.

LB-037

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What assurance statement about building safety and compliance is supportable today, and which material uncertainties must be disclosed alongside it?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: only owner-approved aggregate status from current safety systems, compliance registers and action trackers, with source-health and timestamp flags. Spatial coverage: the exact assets, duties, period and building scope named in the statement. Observation and aggregation: current snapshot plus trend or exception history at the reporting cut-off. Decision-use gate: complete reconciliation with authoritative registers and explicit scope, exclusions, expired evidence and open actions. Known limits: no dashboard can establish all hazards, legal duties, competence or real-world system performance. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: duty holders, governance, health and safety, fire safety, Estates compliance, legal and communications owners. Required records: signed certificates, current registers, risk assessments, assurance check findings, open actions, exemptions, competent advice and approved wording. Source precedence: duty-holder and competent professional statements outrank analytic summaries; material dissent cannot be removed for presentation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: assurance expires at the owner-defined cut-off or on any material change or new impairment. Unresolved conflicts: scope, legal interpretation, evidence age, open actions, exceptions and professional opinion; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: claim-to-evidence assurance synthesis. Decision interval: at each governance report and before external communication. Minimum evidence: precise claim, duty and scope, current signed evidence, open-action status, competent owner and decision cut-off. Method: classify each clause as verified fact, professional judgement, estimate or unknown; test wording against evidence scope and include material qualifications. Quantified uncertainty: state evidence coverage, unresolved conflict, open actions and confidence in the claim rather than a composite green score; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the relevant duty holder, governance secretary and communications owner.

ANSWER BOUNDARY

The service must not infer safety, compliance or acceptable risk beyond the scope of current competent evidence. It must not disclose restricted vulnerabilities, personal case information or privileged legal advice outside authorised roles. It must not promise or certify legal compliance, safe occupation, absence of hazards or future performance; it must not authorise or control public wording, risk acceptance, reopening, remediation or regulatory response. Use only wording approved by accountable owners.

WHY THIS IS USEFUL

Leadership can issue a time-bounded assurance statement that separates verified facts, professional judgement, open actions and unknowns.

LB-038

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which high-consequence risk depends on a single asset, route, supplier, manual step or untested assumption with no verified fallback?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: availability and alarm telemetry for critical assets, utilities, routes, communication links and backup systems, plus test-event status. Spatial coverage: only critical service chains in the approved risk and continuity scope; restricted topology shown to authorised roles. Observation and aggregation: event-driven status, monthly dependency snapshots and exercise or test evidence. Decision-use gate: owner-validated dependency graph, current criticality, tested failover path and known telemetry blind spots. Known limits: human, supplier and process dependencies may not emit telemetry and dormant failures may appear only under demand. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: risk owners, business continuity, procurement, Estates, IT, security, School operations and critical service owners. Required records: risk register, business-impact analysis, supplier contracts, dependency maps, maintenance, exercises, competence and succession arrangements. Source precedence: current risk-owner assessment and tested evidence outrank assumed redundancy or vendor claims. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: dependency and supplier evidence is stale after contract, configuration, staffing, route or service change. Unresolved conflicts: risk register, technical topology, supplier capability, manual workaround and ownership of untested assumptions; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: single-point-of-failure and assumption-challenge analysis. Decision interval: quarterly and before material change. Minimum evidence: defined critical outcome, complete owner-validated dependency chain, consequence threshold and evidence of last successful test for each claimed control. Method: identify nodes with no independent alternative, distinguish designed redundancy from tested end-to-end resilience and rank untested assumptions by consequence. Quantified uncertainty: show dependency completeness, test age, common-cause exposure and confidence that alternatives are independent; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable risk owner and relevant technical, supplier or continuity owner.

ANSWER BOUNDARY

The service must not infer acceptable risk, supplier failure probability or staff capability from incomplete dependency data. It must not disclose restricted vulnerabilities, personal succession details or commercially sensitive supplier information beyond authorised roles. It must not promise or certify resilience, control effectiveness or adequacy of risk treatment; it must not authorise or control risk acceptance, supplier action, staffing, testing or investment. The analysis identifies reliance; it does not approve the response.

WHY THIS IS USEFUL

Risk owners can remove a single point of failure, verify a fallback or accept residual risk through the proper governance route.

LB-039

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are temporary works, contractors, events or decants creating unmanaged interfaces with occupied operations, inclusive access or safety controls?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: approved work-zone access status, temporary power and environmental monitoring, route closures, alarm isolations, lift or door status and anonymous interface counts. Spatial coverage: temporary boundaries and shared interfaces in the current plan; no worker or occupant tracking. Observation and aggregation: event-driven status and 1-5 minute monitoring during active work, with daily exception summaries. Decision-use gate: current permit and boundary map, commissioned temporary sensors, clear handover and immediate update after layout or isolation change. Known limits: telemetry cannot verify method compliance, competence, housekeeping, supervision or every changing site condition. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: project or event sponsor, principal contractor, Estates, School operations, health and safety, security, accessibility and affected service owners. Required records: current permits, method statements, temporary-works design, phasing, isolation register, access routes, decant plan, inspections and handovers. Source precedence: competent site control, permits and current on-site instruction outrank analytics and historical plans. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: temporary information is stale immediately after site, sequence, isolation, route or occupancy change. Unresolved conflicts: contractor plan, permit status, on-site condition, occupied-service needs and ownership at shift or phase handover; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: temporary-interface completeness and exception analysis. Decision interval: before each phase and daily during active high-risk interfaces. Minimum evidence: current plan and permits, named interface owners, verified boundaries, affected-service sign-off and evidence of daily inspection. Method: map each temporary condition to affected controls and services, flag unowned or conflicting interfaces and require physical verification for site condition. Quantified uncertainty: show information age, unmonitored interfaces, inspection gaps and conditions that invalidate the plan; report response count and rate, suppressed cells and an interval or range; do not generalise from convenience samples. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project or event sponsor, competent site manager and affected service owners.

ANSWER BOUNDARY

The service must not infer worker behaviour, competence, compliance or site safety from telemetry. It must not disclose personal access records, security-sensitive work details or confidential occupied activities. It must not promise or certify permit compliance, safe working, safe occupation or adequacy of temporary controls; it must not authorise or control stop-work, isolation, access, event approval or contractor instruction. On-site competent control and official instructions take precedence.

WHY THIS IS USEFUL

Project and operational owners can align permits, segregations, routes, communications and handovers before occupied activity continues.

LB-040

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which current safety or compliance question cannot be answered from competent current evidence and must be escalated rather than simplified?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: source-health, asset-mapping, status and timestamp evidence for only the safety systems or registers cited in the question. Spatial coverage: the exact duty, asset, location and time period under assessment. Observation and aggregation: continuous health where available and a decision-cut-off snapshot. Decision-use gate: authoritative owner, current record, competent scope and explicit confirmation that all material components are represented. Known limits: more telemetry cannot substitute for missing inspection, competence, legal interpretation or physical verification. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: duty holder, compliance manager, health and safety, fire safety, Estates, legal and relevant competent advisers. Required records: question-specific obligation, current records, professional advice, inspection, certificate, action status and known evidence gaps. Source precedence: current competent evidence and duty-holder interpretation outrank any analytic conclusion. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: safety evidence is stale at expiry, material change, new incident, failed test or superseding advice. Unresolved conflicts: record scope, asset identity, professional interpretation, open actions and unresolved ownership; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: safety-claim sufficiency and escalation testing. Decision interval: immediately for material questions and at governance assessment. Minimum evidence: precise claim, identified duty holder, competent evidence scope, current records and a defined consequence of answering incorrectly. Method: test whether every material premise has authoritative and competent support; do not average conflicting evidence; name the missing decision owner or inspection. Quantified uncertainty: state exactly what is unknown, why it matters and whether delay, restriction or interim control requires accountable judgement; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the duty holder and the competent professional for the relevant hazard or obligation.

ANSWER BOUNDARY

The service must not infer safety, compliance, acceptable risk or absence of hazard where competent evidence is missing. It must not disclose restricted vulnerability, personal case or privileged legal material beyond authorised roles. It must not promise or certify legal status, safe occupation or sufficiency of interim measures; it must not authorise or control escalation outcome, closure, restriction, remediation or risk acceptance. The correct service response may be no answer other than a named escalation.

WHY THIS IS USEFUL

The duty holder can stop an unsupported assurance claim and route the question to the competent person or current record owner.

LB-041

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which assets impose the highest combined burden of reactive cost, downtime, condition risk and consequence for priority services?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: asset run-state, alarm and fault telemetry, runtime counters, condition points such as vibration or temperature where justified, and downstream service availability. Spatial coverage: assets with verified identifiers, maintainable boundaries and current links to teaching, research, workplace or public services. Observation and aggregation: event-driven faults, hourly or daily runtime, monthly reliability and annual cost summaries. Decision-use gate: asset identity reconciled across telemetry and finance, complete work-order linkage and at least 12 months of usable history or explicit new-asset limitation. Known limits: unrecorded labour, shared costs, dormant defects and changing duty cycles can distort comparisons. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: asset owners, Estates maintenance, finance, procurement, School operations and affected service owners. Required records: current asset register, condition survey, work orders, labour and parts cost, warranty, criticality, outage consequence and renewal plan. Source precedence: owner-confirmed asset identity and competent condition assessment outrank inferred condition from fault counts. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: condition and cost evidence is stale after major corrective work, duty change, project handover or cost-base version change. Unresolved conflicts: asset identifiers, cost allocation, failure coding, condition grade and service criticality; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: asset risk-cost-performance prioritisation. Decision interval: quarterly and at annual capital planning. Minimum evidence: current asset identity, 12 months of cost and downtime where available, competent condition, criticality and downstream service consequence. Method: normalise costs only within comparable asset classes, separate frequency from consequence and identify assets dominated by repeated reactive intervention. Quantified uncertainty: show incomplete cost attribution, event-count intervals, condition uncertainty and ranking sensitivity to criticality assumptions; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable asset owner, finance partner and affected service owner.

ANSWER BOUNDARY

The service must not infer maintainer performance, negligence or asset condition from alarm frequency alone. It must not disclose personal work records, commercially restricted costs or security-sensitive asset details beyond authorised roles. It must not promise or certify remaining life, failure probability, value for money or need for replacement; it must not authorise or control capital allocation, maintenance strategy, procurement or asset shutdown. Competent condition assessment remains necessary.

WHY THIS IS USEFUL

Asset leaders can focus condition surveys, renewal planning and funding cases on assets with the greatest combined cost and service consequence.

LB-042

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

For which assets is planned renewal now more defensible than continued intervention, once whole-life cost, risk and disruption are considered?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: asset fault, downtime, runtime, service-load, energy and environmental telemetry relevant to each renewal option. Spatial coverage: candidate assets and affected service chains using current asset, room and dependency mappings. Observation and aggregation: monthly baselines, incident histories and scenario summaries over the appraisal horizon. Decision-use gate: comparable asset boundaries, current condition, reconciled costs and sufficient operating history to estimate normal variation. Known limits: future technology, prices, policy, use and hidden condition create substantial forecast uncertainty. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: asset and service owners, Estates strategy, finance, procurement, sustainability, risk, accessibility and School leadership. Required records: condition surveys, maintenance and energy cost, outages, criticality, option scopes, lifecycle assumptions, project dependencies and approved standards. Source precedence: mandatory duties and competent condition evidence constrain options before whole-life optimisation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: cost and condition assumptions are stale after market, scope, failure, service or policy change. Unresolved conflicts: condition, remaining life, cost base, risk consequence, carbon boundary and service benefit; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: whole-life cost, risk and disruption option appraisal. Decision interval: at capital planning and project gateways. Minimum evidence: current condition, baseline cost, service consequence, at least two credible options, lifecycle assumptions, discount basis and residual-risk owner. Method: model cost and disruption ranges, include planned and reactive work, test early or late renewal and expose break-even or trigger conditions. Quantified uncertainty: show scenario ranges, parameter sensitivity, unpriced effects and share of tested cases in which the preferred option changes; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the asset strategy owner, finance partner and accountable service owner.

ANSWER BOUNDARY

The service must not infer precise remaining life, failure timing or institutional preference from a model. It must not disclose commercial bids, restricted vulnerabilities or personal work records beyond authorised roles. It must not promise or certify affordability, value for money, safety, carbon performance or project success; it must not authorise or control renewal, design, funding, procurement or risk acceptance. The analysis informs a gateway; it does not replace professional option development.

WHY THIS IS USEFUL

The asset owner can select continued intervention, planned renewal or interim risk control using transparent whole-life evidence.

LB-043

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are completed capital projects delivering the approved capacity, service, inclusion, carbon and reliability outcomes after handover?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: project-specific availability, occupancy context, environmental, energy, fault, access-feature and service-load telemetry. Spatial coverage: the delivered assets and spaces plus agreed comparison boundaries, all remapped after handover. Observation and aggregation: commissioning snapshots, matched baseline and post-handover periods, monthly trends and defects-period assessments. Decision-use gate: verified commissioning, stable definitions, at least eight comparable weeks before and after where feasible and all known defects flagged. Known limits: ramp-up, seasonal change, incomplete user adoption and concurrent works can obscure benefit realisation. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: project sponsor, project manager, asset owner, School and service benefit owners, finance, accessibility and sustainability. Required records: business case, approved requirements, commissioning and acceptance evidence, as-built records, defects, costs, benefits map and post-occupancy evaluation. Source precedence: approved requirements and competent acceptance evidence outrank analytic proxies or pre-project spatial records. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: as-built, benefit and cost evidence is stale after variation, defect, operating change or rebaseline. Unresolved conflicts: approved scope, delivered asset, commissioning status, baseline, benefit ownership and unresolved defects; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: post-occupancy benefit, reliability and defect analysis. Decision interval: at handover, defects close-out and agreed annual assessments. Minimum evidence: traceable requirements, verified delivered scope, comparable baseline, current defects and named owner for every material benefit. Method: compare actual service with requirements and baseline, distinguish commissioning defect, operational change and unmeasured benefit, and test persistence. Quantified uncertainty: show effect ranges, incomplete baseline, confounders, missing stakeholder evidence and confidence by benefit; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project sponsor and accountable asset or benefit owners.

ANSWER BOUNDARY

The service must not infer project success, user benefit, learning, productivity or research quality from telemetry alone. It must not disclose personal outcomes, commercially restricted defect details or small-group use. It must not promise or certify contract completion, compliance, value for money or causation of benefit; it must not authorise or control acceptance, payment, defect remedy, operational change or future investment. Formal contract and competent acceptance processes remain authoritative.

WHY THIS IS USEFUL

The sponsor can accept the project outcome, require defects or tuning to be addressed, extend measurement or revisit the benefits case.

LB-044

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which gaps in asset identity, ownership, condition, warranty or maintenance history prevent a responsible lifecycle decision?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: telemetry endpoint identity, last-seen status, location mapping, firmware or controller identifier where permitted, runtime and alarm provenance. Spatial coverage: only assets in the lifecycle decision scope, with no discovery of personal devices or unauthorised network content. Observation and aggregation: continuous health and a reconciliation snapshot at each planning cut-off. Decision-use gate: each endpoint linked to a current asset identifier, owner, location, service and maintenance record with duplicate and orphan checks. Known limits: an observed endpoint may not equal a maintainable asset and uninstrumented assets remain invisible. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: asset information management, asset owners, Estates maintenance, finance, procurement, project handover and records management. Required records: current asset register, as-built and O&M information, condition, warranty, ownership, maintenance plan, work history and disposal status. Source precedence: owner-approved asset register and signed handover evidence outrank telemetry names or the May 2021 floor plan. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: records are stale after move, replacement, refurbishment, ownership, warranty or service change. Unresolved conflicts: asset identifiers, location, owner, warranty, status, duplicates, retired assets and incomplete handover; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: asset-record completeness, consistency and decision-fitness assessment. Decision interval: monthly for critical exceptions and before lifecycle planning. Minimum evidence: defined asset population, authoritative identifier, owner, location, service, condition or evidence route and current status for each material asset. Method: reconcile sources, classify missing and conflicting attributes, assess which gaps could change maintenance, warranty or renewal decisions. Quantified uncertainty: show coverage by asset class, unresolved duplicates, confidence in matches and cost or risk exposure of unknown records; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the asset information owner and accountable asset or project handover owner.

ANSWER BOUNDARY

The service must not infer asset existence, condition, ownership or disposal from telemetry alone. It must not disclose security-sensitive configurations, commercial warranty details or personal device information beyond authorised roles. It must not promise or certify register completeness, remaining life or suitability for investment; it must not authorise or control record correction, write-off, maintenance, procurement or disposal. Physical verification may be the smallest adequate remedy.

WHY THIS IS USEFUL

Information and asset owners can close the record gaps that block condition, warranty, ownership and lifecycle decisions.

LB-045

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What recurring financial and service consequences arise from space that is constrained, unsuitable, fragmented or persistently lightly used?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous occupancy state, doorway counts, booking-panel status, environmental and service availability, plus shared-zone energy and maintenance telemetry. Spatial coverage: current space categories aggregated to protect confidential, specialist and small-group use. Observation and aggregation: 5-15 minute operational samples, weekly profiles and annual cost summaries. Decision-use gate: current room purpose, usable capacity, at least two representative terms, 85% valid data and explicit exclusions for specialist and contingency space. Known limits: low occupancy does not establish low value, surplus, suitability or avoidable cost. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School and University space owners, finance, timetabling, research facilities, workplace, accessibility and Estates. Required records: current allocation, functional brief, capacity, booking, maintenance and energy cost, service constraints, strategic need and stakeholder evidence. Source precedence: approved purpose, specialist need and competent suitability assessment outrank occupancy rate and historic room labels. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: space and cost records are stale after use, layout, allocation, tariff or service change. Unresolved conflicts: local and central ownership, cost allocation, occupancy interpretation, specialist need and alternative-space availability; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: space consequence, cost and alternative-use scenario analysis. Decision interval: annually and before estate-option decisions. Minimum evidence: current purpose and owner, representative use, full service-cost range, suitability evidence, constraints and credible alternatives. Method: separate low use, constrained use and unsuitable use; estimate avoidable and unavoidable cost; test sharing, adaptation, consolidation and no-change scenarios. Quantified uncertainty: show missing demand, cost allocation uncertainty, specialist-value evidence and assumptions that drive the option result; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable space owner, finance partner and affected service owners.

ANSWER BOUNDARY

The service must not infer space value, productivity or entitlement from occupancy. It must not disclose who uses a space, research activity, confidential bookings or small-group patterns. It must not promise or certify surplus status, affordability, suitability or benefit of consolidation; it must not authorise or control space release, allocation, relocation, closure or investment. Strategic and specialist value must be represented explicitly.

WHY THIS IS USEFUL

Portfolio leaders can test consolidation, reconfiguration, service changes or protected specialist use against full recurring consequences.

LB-046

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which modest and deliverable work would remove a recurring bottleneck or unlock disproportionate value for several services?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: candidate-specific queue, availability, route, fault, occupancy-context, environmental and service-load indicators. Spatial coverage: the exact bottleneck and affected service interface, with a comparable baseline. Observation and aggregation: 1-15 minute data through representative demand periods, plus incident and work-order history. Decision-use gate: repeated shortfall in at least four representative weeks or verified high-consequence incidents, current mapping and a named owner. Known limits: small works can shift demand, create new interfaces or fail to address hidden causes. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School operations, Estates, accessibility, service owners, finance, procurement and project delivery. Required records: current service issue, condition, cost range, constraints, dependencies, previous corrective works, stakeholder evidence and delivery route. Source precedence: mandatory controls and competent technical or access advice constrain proposed quick wins. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: cost and scope are stale after site condition, market or dependency change. Unresolved conflicts: observed bottleneck, reported experience, technical diagnosis, cost, ownership and interaction with planned projects; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: small-intervention option appraisal and reversible-trial design. Decision interval: quarterly and at minor-works programme assessment. Minimum evidence: verified recurring problem, defined service outcome, competent feasibility check, cost range, dependencies and measurable success and stop criteria. Method: compare no-change, operational and small-works options; estimate effect range; prefer a reversible trial where causation or demand is uncertain. Quantified uncertainty: show baseline variability, implementation uncertainty, spillover risk and share of tested cases in which the intervention meets the stated threshold; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable service owner and minor-works programme owner.

ANSWER BOUNDARY

The service must not infer that a small intervention is low risk or automatically high value. It must not disclose personal feedback, commercial estimates or security-sensitive weaknesses beyond authorised roles. It must not promise or certify deliverability, value for money, safety, accessibility or lasting benefit; it must not authorise or control work approval, funding, procurement, service change or risk acceptance. No quick-win label should bypass normal competent checks.

WHY THIS IS USEFUL

The budget holder can commission a small high-leverage scheme with clear acceptance measures and limited delivery risk.

LB-047

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are outsourced building services meeting outcome-based requirements without hidden backlog, under-reporting or transfer of cost and risk?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: contract-relevant aggregate service availability, response timestamps, demand context, asset status, environmental conditions and quality proxies approved by the owner. Spatial coverage: only services and locations in the contract boundary; no individual worker tracking or content inspection. Observation and aggregation: event and daily operational data, monthly contract summaries and exception-level evidence. Decision-use gate: independent source mapping, agreed definitions, complete denominator, assurance check trail and reconciliation with sampled inspections. Known limits: telemetry can be optimised to a metric, omit unreported work and miss quality, dignity or downstream cost. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: contract owner, procurement, finance, Estates, School operations, health and safety, accessibility and contractor. Required records: current contract, service levels, exclusions, variations, inspection, work orders, invoices, complaints, incidents and improvement actions. Source precedence: contract terms, competent inspections and verified outcome evidence outrank contractor dashboards or isolated sensor metrics. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: measures are stale after variation, service-boundary, price or operating-context change. Unresolved conflicts: contractor returns, independent evidence, exception coding, cost allocation and responsibility for downstream consequence; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: contract-outcome triangulation and gaming-risk analysis. Decision interval: monthly and at formal contract assessment. Minimum evidence: current contract definitions, independent evidence, inspection sample, complete denominator, variations and a named owner for each exception. Method: compare reported and independently observed outcomes, inspect discontinuities near thresholds, test denominator exclusions and track cost or risk transfer. Quantified uncertainty: show inspection sampling error, unreported demand, disputed definitions and confidence in each exception rather than a single vendor score; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the contract owner, procurement lead and relevant competent service owner.

ANSWER BOUNDARY

The service must not infer individual worker performance, misconduct or intent from service metrics. It must not disclose personal workforce data, commercially confidential information or security-sensitive details beyond authorised roles. It must not promise or certify contract compliance, fraud, quality or value for money from analytics alone; it must not authorise or control payment, penalty, contract remedy, staffing or procurement action. Formal contract management and fair investigation remain necessary.

WHY THIS IS USEFUL

The contract owner can challenge performance, enforce evidence requirements, reset incentives or change service design without exposing individual workers.

LB-048

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How much operational, safety, financial and carbon exposure is accumulating through deferred maintenance and obsolescence?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: fault, alarm, runtime, energy, environmental and service-availability telemetry for deferred assets and dependent services. Spatial coverage: assets on the approved deferred-work list with current condition, ownership and dependency mapping. Observation and aggregation: monthly trends, incident summaries and scenario snapshots over the planning horizon. Decision-use gate: reconciled deferral scope, current condition, verified baseline and explicit blind spots for uninstrumented or dormant failure modes. Known limits: failure timing, hidden condition, spare availability and future prices are uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: asset owners, maintenance, finance, risk, safety, sustainability, procurement and affected service owners. Required records: deferred-maintenance register, condition, work orders, inspections, criticality, risk controls, cost estimates, spares and lifecycle plan. Source precedence: current competent safety and condition evidence constrains financial optimisation; formal risk ownership outranks analytic scoring. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: exposure estimates are stale after new failure, inspection, corrective work, cost, policy or service change. Unresolved conflicts: deferral rationale, condition, cost, risk consequence, carbon effect, spare availability and ownership; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: deferred-work exposure and scenario analysis. Decision interval: quarterly and at budget planning. Minimum evidence: current deferral list, competent condition, affected service, cost range, risk owner, current controls and at least no-change and intervention scenarios. Method: estimate ranges for reactive cost, downtime, energy penalty and consequence; separate known exposure from modelled failure scenarios and identify trigger breaches. Quantified uncertainty: show scenario assumptions, residual risk, unpriced consequence and share of tested cases in which rankings change under plausible failure rates; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable asset and risk owners with the finance partner.

ANSWER BOUNDARY

The service must not infer exact failure probability, safe remaining life or acceptable risk from trend data. It must not disclose restricted vulnerabilities, commercial estimates or personal work records beyond authorised roles. It must not promise or certify safety, affordability, carbon saving or value of continued deferral; it must not authorise or control risk acceptance, budget, maintenance, shutdown or renewal. Competent condition and safety advice remain decisive.

WHY THIS IS USEFUL

Leadership can see the cost and service consequence of delay, set funded priorities and document any consciously retained exposure.

LB-049

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which project option remains preferred when cost, mission benefit, risk, carbon, inclusion and disruption assumptions are varied?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: option-relevant aggregate service, demand, asset, energy, environmental, access-feature and outage indicators with common baseline boundaries. Spatial coverage: only assets, spaces and services within each defined option and its consequential impacts. Observation and aggregation: baseline summaries, construction or transition scenarios and lifecycle model inputs. Decision-use gate: comparable boundaries and periods, current mappings, documented missingness and no metric included without an interpretable decision link. Known limits: many benefits and harms remain qualitative, distributional or dependent on future behaviour and policy. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: project sponsor, finance, Estates strategy, School leadership, risk, safety, sustainability, accessibility, procurement and affected service owners. Required records: approved objectives, constraints, option scopes, cost plan, programme, risk, carbon assessment, access appraisal, stakeholder evidence and dependencies. Source precedence: mandatory duties and competent advice eliminate or constrain options before discretionary trade-offs. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: options are stale after scope, price, design, policy, programme or service-assumption change. Unresolved conflicts: option versions, boundaries, costs, benefits, risk, carbon, inclusion and affected-owner judgements; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: transparent multi-criteria decision analysis with scenario and sensitivity testing. Decision interval: at option selection and each project gateway. Minimum evidence: at least two feasible options plus no-change, common criteria, owner-approved ranges, explicit constraints, distributional impacts and unresolved dependencies. Method: screen constraints, compare ranges rather than point scores, test alternative weights and scenarios, and display non-compensable duties separately. Quantified uncertainty: show parameter ranges, rank reversals, unquantified effects and assumptions that dominate the result; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project sponsor and accountable owners of each material criterion.

ANSWER BOUNDARY

The service must not infer that a composite score determines the institutional choice or that unlike harms are fully compensable. It must not disclose commercially restricted estimates, privileged advice, personal feedback or vulnerabilities beyond authorised roles. It must not promise or certify value for money, affordability, compliance, safety, carbon outcome or project success; it must not authorise or control option selection, funding, procurement, scope or risk acceptance. Accountable judgement and professional advice must remain visible.

WHY THIS IS USEFUL

The sponsor can select an option whose ranking remains stable, or identify the assumption that must be resolved before commitment.

LB-050

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What evidence is still needed to choose responsibly among retaining, refurbishing, repurposing, consolidating or replacing a space or asset?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current condition, fault, downtime, demand context, service load, energy, environmental and access-feature telemetry relevant to the option. Spatial coverage: the asset or space and all material dependencies or receiving locations in each scenario. Observation and aggregation: multi-term or annual baseline, incident history and option-scenario summaries. Decision-use gate: current mapping and condition, comparable cost and service boundaries, sufficient history for recurring demand and explicit treatment of specialist or resilience use. Known limits: future demand, hidden condition, relocation effects and strategic value remain uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accountable asset or space owner, School and University planning, finance, Estates, risk, sustainability, accessibility and affected service owners. Required records: current ownership and purpose, condition, costs, demand, strategic need, duties, dependencies, option studies and stakeholder evidence. Source precedence: approved strategic need, mandatory duties and competent condition or suitability evidence outrank utilisation or cost alone. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: evidence is stale after condition, demand, ownership, option, cost or policy change. Unresolved conflicts: strategic purpose, use interpretation, condition, cost, receiving-space capacity, carbon and stakeholder consequence; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: portfolio-option appraisal with no-change and transition scenarios. Decision interval: at estate strategy and capital gateways. Minimum evidence: current purpose and owner, competent condition, service consequence, credible options, receiving capacity, cost and carbon ranges, duties and transition plan. Method: compare lifecycle and transition effects, include temporary disruption and stranded risk, and identify trigger conditions rather than force a single recommendation. Quantified uncertainty: show scenario ranges, unmeasured value, dependency uncertainty and whether the option order is robust; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable asset or space owner and University estate-strategy governance.

ANSWER BOUNDARY

The service must not infer that low use, high cost or poor condition alone proves the correct estate option. It must not disclose confidential research activity, personal use, commercial valuations or restricted risks beyond authorised roles. It must not promise or certify affordability, value, safety, deliverability or success of any estate option; it must not authorise or control retention, disposal, consolidation, funding, relocation or procurement. The analysis must preserve strategic, specialist, inclusion and resilience value.

WHY THIS IS USEFUL

The owner can define the next decision gate and evidence commission for retention, refurbishment, repurposing, consolidation or replacement.

LB-051

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Is current energy use proportionate to verified service hours, weather, occupancy context and specialist operational requirements?**SENSORS AND TELEMTRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: fiscal and sub-meter electricity, gas and heat data, plant runtime and setpoints, representative occupancy context, internal conditions and local weather. Spatial coverage: whole building and verified end-use or zone boundaries with current meter hierarchy. Observation and aggregation: 15-30 minute energy and weather data, daily profiles and monthly reporting. Decision-use gate: meter commissioning and reconciliation, clock alignment, at least 12 months of valid coverage where available and documented estimated readings. Known limits: occupancy is an imperfect service proxy and specialist loads, tariffs and shared meters limit comparison. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: energy and sustainability teams, Estates engineering, utility and meter owners, finance, School operations and specialist facility owners. Required records: utility bills, meter register, tariffs, floor area, service hours, weather, plant schedules, specialist-load declarations and approved carbon factors. Source precedence: fiscal meters and current owner-approved meter mapping outrank BMS estimates; approved specialist requirements constrain efficiency claims. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: meter, tariff and carbon-factor data is stale after replacement, boundary, contract or reporting-method change. Unresolved conflicts: fiscal and submeters, shared loads, service hours, specialist requirements and estimated consumption; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: weather- and service-normalised energy performance analysis. Decision interval: monthly with annual assessment. Minimum evidence: reconciled meter boundary, 12 months where feasible, current service hours, weather, material specialist loads and valid data above 90%. Method: construct day-type and seasonal baselines, normalise only supported variables, compare residual demand and investigate persistent deviations. Quantified uncertainty: show model error, unexplained load, missing end uses, confidence intervals and sensitivity to occupancy and weather assumptions; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the energy manager, relevant engineering owner and specialist facility owner.

ANSWER BOUNDARY

The service must not infer waste, staff behaviour, productivity or unjustified activity from energy intensity. It must not disclose individual or small-group occupancy, research activity or commercially restricted tariff details beyond authorised roles. It must not promise or certify energy efficiency, carbon compliance, avoidable use or causation; it must not authorise or control plant control, operating hours, research activity, budgets or targets. Service and safety requirements remain explicit constraints.

WHY THIS IS USEFUL

Energy and service owners can target abnormal demand while protecting verified service hours and specialist requirements.

LB-052

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which zones or assets show persistent avoidable energy or carbon demand without a confirmed service need?**SENSORS AND TELEMTRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: sub-meter energy, circuit or plant runtime, valve or damper state, temperature and occupancy context, and after-hours base load. Spatial coverage: mapped end uses, zones and assets with owner-confirmed service purpose. Observation and aggregation: 5-30 minute data, after-hours profiles, weekly persistence and seasonal summaries. Decision-use gate: current meter-to-asset mapping, at least eight comparable weeks, 90% valid data and owner confirmation of legitimate standby or specialist operation. Known limits: shared meters, control interactions and hidden service requirements can make apparent waste misleading. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: energy manager, Estates engineering, asset owners, School operations, research facilities and sustainability. Required records: meter hierarchy, plant schedules, BMS configuration, service requirements, maintenance, exceptions, tariffs and carbon factors. Source precedence: owner-confirmed service need and safety or research requirements outrank an analytic avoidable-load label. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: service requirements and control schedules are stale after operational, project or seasonal change. Unresolved conflicts: meter attribution, control intent, legitimate standby, maintenance state and responsibility for adjustment; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: persistent-baseload and schedule-mismatch detection. Decision interval: weekly operationally and monthly for management. Minimum evidence: verified end-use mapping, eight comparable weeks, defined service window, owner-confirmed exceptions and a stable baseline. Method: detect repeated load outside required windows, compare with plant state and occupancy context, and estimate a bounded avoidable range rather than all residual load. Quantified uncertainty: show meter allocation error, model range, unknown service need and confidence that the pattern persists; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the energy manager and accountable engineering or service owner.

ANSWER BOUNDARY

The service must not infer waste, negligence or individual behaviour from energy patterns. It must not disclose person-level occupancy, research process details or restricted control configurations. It must not promise or certify avoidable consumption, carbon saving, safety or acceptability of shutdown; it must not authorise or control plant schedules, setpoints, isolation, service hours or targets. Physical and service-owner verification is required before intervention.

WHY THIS IS USEFUL

Estates can investigate persistent base loads, schedules, controls or faulty assets and confirm savings after action.

LB-053

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which claimed carbon savings remain credible after baseline quality, weather, service level and concurrent changes are tested?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: measure-specific fiscal or sub-meter energy, plant runtime, weather, occupancy context, service conditions and implementation status. Spatial coverage: intervention and comparison boundaries with unchanged or fully documented meter remapping. Observation and aggregation: matched pre- and post-implementation periods, monthly reporting and persistence follow-up. Decision-use gate: verified implementation date, at least 12 months pre and post where seasonality demands it, 90% valid data and recorded concurrent changes. Known limits: rebound, service change, tariff change and unmeasured operational intervention can bias savings. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: sustainability, energy, Estates engineering, finance, project sponsor and affected service owners. Required records: approved carbon method, baseline, meter register, carbon factors, project scope, commissioning, costs, service requirements and implementation record. Source precedence: approved reporting method and fiscal data outrank convenience baselines; service and safety requirements constrain interpretation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: baseline and factors are stale after boundary, method, service, meter or factor version change unless formally restated. Unresolved conflicts: implementation completeness, baseline choice, factors, service change, cost and attribution; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: measurement and verification using matched baseline or change-point models. Decision interval: at benefits assessments and annual carbon reporting. Minimum evidence: approved boundary and method, valid pre/post data, implementation evidence, weather and service covariates, and concurrent-change record. Method: estimate adjusted savings, test pre-trend and persistence, compare alternative baselines and reconcile with fiscal totals. Quantified uncertainty: report savings interval, model error, factor uncertainty, residual confounding and claim grade; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the energy and sustainability owners with the project benefit owner.

ANSWER BOUNDARY

The service must not infer causation or permanent saving from a raw before-after difference. It must not disclose individual occupancy, restricted operational details or commercially confidential costs beyond authorised roles. It must not promise or certify carbon compliance, future savings, payback or project success; it must not authorise or control targets, plant operation, funding, public claim or rollout. Restatements and material uncertainty must remain visible.

WHY THIS IS USEFUL

Leadership can report only defensible savings, correct weak claims and decide whether further measurement or tuning is warranted.

LB-054

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which energy-saving controls can remain in place without reducing verified comfort, ventilation or indoor-air-quality performance below current requirements?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: representative temperature, humidity, CO2 and where justified particulate sensors, ventilation flow or fan status, setpoints, valve or damper state and energy. Spatial coverage: occupied zones and activity-specific spaces with professionally selected sensor locations. Observation and aggregation: 1-5 minute environmental and plant data, hourly distributions and daily energy summaries. Decision-use gate: documented siting and calibration, current room use, 90% valid data and owner-approved comfort or ventilation criteria. Known limits: CO2 is a proxy, spatial variation is incomplete and health or infection risk cannot be inferred directly. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: Estates engineering, energy, health and safety, occupational health, research facilities, accessibility and room owners. Required records: current design and operating criteria, risk assessments, ventilation tests, maintenance, complaints, energy baseline and optimisation change records. Source precedence: competent ventilation and activity-specific requirements outrank energy optimisation targets or generic comfort bands. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: criteria are stale after use, occupancy, ventilation, process or guidance change. Unresolved conflicts: sensor readings, plant control, direct experience, activity requirements and interpretation of acceptable conditions; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: joint service-condition and energy constraint analysis. Decision interval: weekly during optimisation and monthly thereafter. Minimum evidence: current criteria, calibrated sensors, valid plant data, representative operating periods, control-change record and named service owner. Method: compare conditions and energy before and after changes, quantify excursions and identify whether saving occurs inside or outside approved service constraints. Quantified uncertainty: show sensor tolerance, spatial uncertainty, missing periods, model error and cases requiring competent investigation; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the engineering owner, health-and-safety or occupational-health adviser and energy manager.

ANSWER BOUNDARY

The service must not infer individual comfort, health, exposure, productivity or infection risk from room telemetry. It must not disclose personal health information, individual occupancy or confidential activity details. It must not promise or certify safe air, compliance, universal comfort or suitability of an energy-saving control; it must not authorise or control setpoints, ventilation, room use, closure or work restrictions. Safety and competent environmental requirements are non-negotiable constraints.

WHY THIS IS USEFUL

Energy actions can proceed only where comfort, ventilation and indoor-air-quality evidence remains within current requirements.

LB-055

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Where does the current end-to-end accessible journey fail across entrances, routes, rooms, amenities and emergency arrangements?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: lift, powered-door, refuge-communication, visual-alert, route and amenity availability, plus optional anonymous barrier-category feedback. Spatial coverage: complete currently mapped journeys to priority services, without tracking a person or inferring disability. Observation and aggregation: event-driven status, daily journey-availability summaries and termly barrier patterns with small-count suppression. Decision-use gate: asset-route mapping physically validated, event delivery above 95% and current professional access assessment for any assurance claim. Known limits: telemetry cannot capture effort, dignity, sensory or cognitive barriers, assistance quality or individual suitability. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: accessibility and inclusion, Estates, fire safety, School operations, service owners and affected stakeholders through protected engagement. Required records: current access assurance check, route and room information, evacuation arrangements, service tickets, adjustments governance, complaints and project actions. Source precedence: lived experience and current competent access assessment outrank sensor-derived journey availability; the May 2021 plan is historical only. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: journey evidence is stale after route, service, layout, staffing or communication change. Unresolved conflicts: asset status, current access information, personal arrangements, reported experience and professional findings; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: end-to-end journey dependency and barrier analysis. Decision interval: termly and before major change or event. Minimum evidence: current journey map, verified asset status, competent assessment, safe engagement evidence and owner for every dependency. Method: evaluate the whole journey rather than isolated features, identify failed or fragile links and distinguish verified, reported and unknown barriers. Quantified uncertainty: show evidence age, unobserved barrier types, suppressed evidence and confidence in each journey segment; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accessibility lead and accountable owners of the affected route and service.

ANSWER BOUNDARY

The service must not infer disability, capability, need, destination or experience of an individual. It must not disclose adjustment cases, personal evacuation plans, free text or small-group journeys. It must not promise or certify equality compliance, universal accessibility or equivalence of an alternative; it must not authorise or control adjustments, route changes, service access, capital work or emergency arrangements. Direct engagement and competent assessment remain essential.

WHY THIS IS USEFUL

Access and Estates owners can commission an end-to-end barrier assessment and prioritize changes that restore complete usable journeys.

LB-056

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are the benefits, costs and disruptions of building investment being distributed as approved across affected communities and activities?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: project-specific aggregate service availability, access-feature status, demand context, environmental conditions, disruption duration and energy use. Spatial coverage: affected services and locations grouped at a safe level; no demographic sensing or individual outcome tracking. Observation and aggregation: matched baseline, delivery and post-handover periods with milestone summaries. Decision-use gate: current affected-group map from authorised owners, safe aggregation, verified project boundaries and at least one direct engagement source for each material group. Known limits: sensor evidence cannot establish fairness, dignity, lived impact or all indirect costs. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: project sponsor, equality and inclusion, School and University planning, finance, service owners, sustainability and stakeholder representatives. Required records: approved equality or distributional assessment, benefit map, cost and disruption plan, engagement, complaints, access arrangements and delivered scope. Source precedence: protected engagement and competent equality assessment outrank telemetry or aggregate usage. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: affected-group and distribution evidence is stale after scope, phasing, population or service change. Unresolved conflicts: planned and actual benefit, disruption, direct and indirect cost, missing voices and ownership of mitigation; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: distributional benefit and burden assessment. Decision interval: at option selection, delivery gateways and benefits assessment. Minimum evidence: defined affected groups and services, authorised evidence, safe aggregation, project baseline, mitigation commitments and named owners. Method: compare intended and observed distribution, distinguish benefit, burden, delay and unmeasured effect, and preserve qualitative evidence that cannot be monetised. Quantified uncertainty: show missing groups, response bias, suppressed cells, uncertainty in indirect effects and whether conclusions change under alternative value weights; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the project sponsor, equality lead and affected service owners.

ANSWER BOUNDARY

The service must not infer protected characteristics, fairness or individual benefit from location or aggregate use. It must not disclose personal outcomes, case details, free text or small groups. It must not promise or certify equality compliance, equitable distribution, social value or absence of harm; it must not authorise or control mitigation, scope, funding, engagement or project approval. Low-volume high-consequence impacts must remain visible.

WHY THIS IS USEFUL

The sponsor can change phasing, engagement, mitigation or benefit allocation where distribution differs from the approved case.

LB-057

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Can the building reliably support the access, facilities and service standards required by approved civic and public-engagement commitments?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: anonymous visitor entrance counts, route and lift status, room and amenity availability, queue aggregates, environmental conditions and service faults. Spatial coverage: public-facing routes, rooms and services named in the current engagement plan. Observation and aggregation: event-level and monthly summaries, with pre-event readiness and post-event disruption snapshots. Decision-use gate: current event and route mapping, verified service status, safe aggregation and direct partner or participant feedback for experience claims. Known limits: counts do not capture who is excluded, trust, engagement quality or community benefit. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: School public-engagement lead, University civic mission, event sponsor, accessibility, security, Estates and partner organisations. Required records: approved engagement commitments, event plans, access information, service standards, partner feedback, incidents and corrective actions. Source precedence: partner and participant evidence plus current competent access and safety advice outrank attendance volume. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: plans and access statements are stale after route, service, event or partner change. Unresolved conflicts: commitment scope, attendance, partner experience, service reliability and responsibility for barriers; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: commitment-to-service readiness and delivery analysis. Decision interval: before each major event and annually for programme assessment. Minimum evidence: defined commitment, current event requirements, verified route and service status, partner evidence and owner for corrective action. Method: map commitments to required capabilities, compare planned and actual availability, and distinguish reach, reliability, access and unmeasured civic outcome. Quantified uncertainty: show event variability, missing voices, service blind spots and limits of attendance as an outcome measure; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the public-engagement owner and accountable building or event service owners.

ANSWER BOUNDARY

The service must not infer community benefit, inclusion, trust or engagement quality from visitor counts. It must not disclose participant identities, partner confidential information or small-community feedback without permission. It must not promise or certify delivery of civic outcomes, universal access or absence of reputational risk; it must not authorise or control event approval, access, funding, public claim or partnership decision. Partner and community interpretation must remain visible.

WHY THIS IS USEFUL

Civic and building owners can protect the facilities and reliability needed for approved public commitments or reset commitments honestly.

LB-058

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are procurement and contractor social-value commitments specific, evidenced and attributable without exposing individuals or overstating causation?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: only approved aggregate service, resource, travel, waste, energy or delivery telemetry directly linked to a contractual social-value output. Spatial coverage: the contract and delivery locations in scope, with no worker, learner or community-member tracking. Observation and aggregation: monthly or milestone summaries aligned to contract reporting periods. Decision-use gate: stable definitions, independent provenance, complete denominator and reconciliation with sampled documentary evidence. Known limits: many social outcomes require qualitative, longitudinal and community evidence not available from building sensors. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: procurement and contract owner, supplier, sustainability, equality, finance, legal and community or skills programme owners. Required records: contract commitments, method statements, supplier returns, invoices, verified beneficiary aggregates, assurance check evidence and approved impact definitions. Source precedence: contract terms and independently verified evidence outrank supplier narrative or convenient activity counts. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: commitments and baselines are stale after variation, scope, method or reporting-period change. Unresolved conflicts: definition, attribution, double counting, beneficiary evidence, commercial claims and owner responsibility; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: commitment verification, attribution and double-count screening. Decision interval: at contract milestones and annual reporting. Minimum evidence: specific commitment, unit, baseline, delivery owner, source provenance, independent check and rule preventing duplicate attribution. Method: separate inputs, activities, outputs and outcomes; verify contractual linkage; grade attribution strength and preserve unmeasured community effects. Quantified uncertainty: show coverage, sampling error, attribution limits, counterfactual uncertainty and claims supported only by supplier assertion; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the procurement contract owner and relevant social-value outcome owner.

ANSWER BOUNDARY

The service must not infer individual benefit, protected characteristic, employment quality or community impact from aggregate delivery counts. It must not disclose personal beneficiary or workforce data, commercially confidential detail or free text beyond permission. It must not promise or certify social impact, additionality, contract compliance or value for money; it must not authorise or control payment, contract remedy, public claim, supplier selection or programme continuation. Do not convert activity volume into unsupported outcome claims.

WHY THIS IS USEFUL

Procurement can verify delivery, strengthen contract evidence and report social value without attributing outcomes beyond demonstrated contribution.

LB-059

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How do whole-life cost, operational energy, embodied carbon, maintenance, replacement cycles and end-of-life effects compare across options?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current energy and runtime baselines, asset condition and replacement history, service demand and project-specific operational assumptions. Spatial coverage: common functional unit and service boundary for each option, including enabling works and displaced assets. Observation and aggregation: historic monthly energy and lifecycle scenario inputs over the appraisal horizon. Decision-use gate: reconciled meter boundary, verified quantities where available, declared data-quality class and no omission of material lifecycle stage. Known limits: future grid factors, service use, maintenance, product declarations and end-of-life routes remain uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: sustainability, project sponsor, Estates, quantity surveying, asset owners, procurement, finance and design team. Required records: approved carbon method, quantities, environmental product declarations, energy baseline, maintenance plan, service life, replacement and end-of-life scenarios. Source precedence: approved method and verified project quantities outrank generic product claims or incomplete databases. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: carbon evidence is stale after design, quantity, supplier, factor, service-life or method change. Unresolved conflicts: functional unit, boundary, product data, service life, maintenance, grid factors and treatment of uncertainty; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: whole-life carbon option analysis with scenario and sensitivity testing. Decision interval: at concept, design and procurement gateways. Minimum evidence: common functional unit, complete material boundary, current quantities, operational baseline, service-life assumptions and data-quality rating. Method: calculate ranges by lifecycle stage, test grid and service-life scenarios, identify carbon break-even and prevent double counting of avoided replacement. Quantified uncertainty: show data-quality classes, parameter ranges, omitted stages and share of tested cases in which option order changes; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the sustainability owner, project sponsor and competent cost or design advisers.

ANSWER BOUNDARY

The service must not infer precise future carbon, environmental superiority or service equivalence from model outputs. It must not disclose commercial supplier data beyond permission or restricted project details. It must not promise or certify carbon neutrality, compliance, whole-life accuracy or project value; it must not authorise or control option selection, specification, procurement, funding or offsetting. Safety, inclusion and service requirements cannot be traded away by a carbon score.

WHY THIS IS USEFUL

The project team can compare options on a common lifecycle boundary and expose when carbon or cost conclusions depend on uncertain assumptions.

LB-060

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which coordinated measures could improve carbon, inclusion and service value together without weakening safety or minimum standards?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: action-specific energy, environmental, access-feature, service-availability, demand-context and fault telemetry. Spatial coverage: the combined set of affected assets, spaces, routes and services, including interactions between actions. Observation and aggregation: representative baselines, trial windows and monthly outcome summaries. Decision-use gate: verified mappings, approved minimum conditions, at least one direct inclusion evidence source and explicit monitoring of adverse service effects. Known limits: synergies and trade-offs may emerge only after implementation and some inclusion value is not sensor-observable. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: sustainability, accessibility, health and safety, Estates engineering, School operations, finance and affected service owners. Required records: current standards and duties, carbon plan, access assessments, service levels, maintenance, cost, stakeholder evidence and candidate action scopes. Source precedence: safety, accessibility and other mandatory minimums are constraints, not weights that can be offset by carbon saving. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: action evidence is stale after service, design, policy, user or operating-context change. Unresolved conflicts: carbon, inclusion, comfort, reliability, cost, owner capacity and interaction between measures; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: constraint-led multi-objective option and trial analysis. Decision interval: quarterly and at programme gateways. Minimum evidence: defined actions, current minimum standards, verified baselines, owners, cost ranges, interaction map and success and stop criteria. Method: remove options breaching minimums, identify synergies and conflicts, compare bundles and prefer reversible trials where interaction is uncertain. Quantified uncertainty: show ranges, non-compensable effects, rank sensitivity, distributional uncertainty and conditions triggering rollback; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the programme owner and competent safety, access and engineering owners.

ANSWER BOUNDARY

The service must not infer that a combined score makes a trade-off acceptable or that all stakeholder value is measurable. It must not disclose personal needs, protected case data, commercial details or vulnerabilities beyond authorised roles. It must not promise or certify safety, compliance, equality, carbon outcome or value for money; it must not authorise or control programme, trial, plant control, funding, access or risk acceptance. No action may dilute mandatory protections.

WHY THIS IS USEFUL

Leadership can approve an integrated package with explicit safeguards, owners and acceptance criteria instead of isolated trade-offs.

LB-061

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which building services are genuinely mission-critical, who owns each dependency, and which upstream failure could stop delivery?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: availability and alarm telemetry for power, water, ventilation, network, controls, access, lifts, environmental services and critical shared equipment. Spatial coverage: only services in the approved business-impact scope, with restricted topology visible only to authorised roles. Observation and aggregation: event-driven status, daily availability and quarterly dependency snapshots. Decision-use gate: owner-validated service and asset mapping, tested signal paths, current criticality and a named owner for every dependency edge. Known limits: human, supplier, data and process dependencies may be uninstrumented or only fail under stress. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: business continuity, School leadership, Estates, IT, security, research facilities, procurement and each critical service owner. Required records: business-impact analysis, service catalogue, dependency maps, recovery objectives, contracts, maintenance and current continuity plans. Source precedence: current owner-approved impact and dependency records outrank inferred criticality or historical spatial information. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: criticality and dependencies are stale after service, technology, supplier, occupancy or organisational change. Unresolved conflicts: service ownership, upstream dependencies, recovery objectives, supplier assumptions and cross-team boundaries; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: critical-service ownership and dependency-graph analysis. Decision interval: quarterly and before material change. Minimum evidence: approved priority outcomes, owner-confirmed critical services, current dependency graph, recovery objectives and evidence for each claimed alternative. Method: map outcomes to services and upstream nodes, flag unowned dependencies, single points and conflicting recovery objectives, and grade each edge by evidence. Quantified uncertainty: show graph completeness, untested assumptions, common-cause exposure and confidence by dependency; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the business-continuity lead and accountable owner of each critical service.

ANSWER BOUNDARY

The service must not infer institutional priority, acceptable outage or resilience from technical availability alone. It must not disclose restricted topology, security vulnerabilities, personal succession details or commercially sensitive contracts beyond authorised roles. It must not promise or certify complete dependency coverage, service resilience or adequacy of continuity plans; it must not authorise or control criticality, ownership, recovery objective, testing or investment. Owners must confirm both mission consequence and technical dependency.

WHY THIS IS USEFUL

Continuity owners can confirm critical services, map upstream dependencies and assign protection and recovery ownership.

LB-062

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How long can priority activities continue through loss of power, water, network, ventilation, access or another utility under current arrangements?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: utility status, UPS or generator load and runtime, tank or pressure level where applicable, network availability, environmental drift and access-system state. Spatial coverage: priority service chains and contingency zones in the current continuity plan. Observation and aggregation: seconds to 5 minute incident data, runtime estimates and periodic test summaries. Decision-use gate: current capacities, fuel or consumable status, tested load, calibrated levels, verified dependency map and owner-confirmed operating limits. Known limits: runtime depends on actual load, hidden common causes, manual action, weather, replenishment and equipment condition. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: business continuity, Estates engineering, IT, security, research facilities, procurement and priority-activity owners. Required records: recovery time and point objectives, backup design, fuel and supplier arrangements, test results, critical loads, procedures and current constraints. Source precedence: tested current capability and competent engineering limits outrank design rating or vendor claim. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: runtime evidence is stale after load, fuel, battery, configuration, maintenance or supplier change. Unresolved conflicts: designed and tested capacity, actual load, manual procedures, replenishment and priority allocation; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: continuity endurance and depletion scenario analysis. Decision interval: at each test, material change and annual continuity assessment. Minimum evidence: current priority load, tested backup capacity, consumable state, environmental limit, manual dependencies and at least low, central and high demand scenarios. Method: simulate depletion and condition drift, include startup and transfer failure, and state the earliest trigger requiring relocation, shutdown or resupply decision. Quantified uncertainty: show runtime ranges, test age, unobserved loads, common-cause assumptions and confidence in each contingency stage; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the relevant utility owner and business-continuity lead.

ANSWER BOUNDARY

The service must not infer guaranteed runtime, safe continued operation or priority of one activity from model output. It must not disclose security-sensitive backup capacities, restricted research dependencies or personal assistance arrangements beyond authorised roles. It must not promise or certify continuity, equipment protection, safety or adequacy of fuel and supplies; it must not authorise or control load shedding, shutdown, relocation, resupply or incident action. Incident command and competent technical advice take precedence.

WHY THIS IS USEFUL

Leadership can set evidence-based continuity thresholds, manual fallback periods and relocation or suspension triggers.

LB-063

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What is the verified impact of the present disruption, and which choices remain reserved to incident command or accountable service owners?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current power, network, water, ventilation, access, lift, alarm and room-availability status plus anonymous aggregate demand where safe. Spatial coverage: only affected services and verified impact areas; no personal tracking or speculative extension beyond current mapping. Observation and aggregation: event-driven data with one-minute or service-appropriate heartbeat and time-stamped incident snapshots. Decision-use gate: current health, aligned clocks, authoritative incident boundary and explicit unknown state for every unverified service. Known limits: telemetry may lag physical conditions, miss people needing help and cannot establish hazard or safe access. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: incident commander, emergency coordinator, Estates, IT, security, communications, School operations and affected service owners. Required records: official incident log, current command decisions, service-owner statements, closures, work orders, continuity actions and communications. Source precedence: incident command, on-scene instruction and official communications outrank analytics and historical information. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: incident status is stale at the next command or field update; all snapshots need a clear cut-off. Unresolved conflicts: command information, telemetry, service-owner status, affected area and unresolved or changing conditions; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: incident service-impact synthesis and uncertainty classification. Decision interval: continuous during the incident at command-approved cadence. Minimum evidence: current incident boundary, authoritative status for each critical service, timestamp, source owner and permission to share. Method: separate confirmed, probable, unverified and restored states; map only verified dependencies; show last update and required owner for each unknown. Quantified uncertainty: state data age, unmonitored zones, conflicting reports and confidence without predicting incident development; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to incident command and the accountable owner of each affected service.

ANSWER BOUNDARY

The service must not infer hazard, safety, occupant status, cause, intent or future incident progression from telemetry. It must not disclose personal location, casualty, assistance, security or vulnerability information beyond authorised roles. It must not promise or certify safe access, all-clear, service restoration or completeness of situational awareness; it must not authorise or control evacuation, cordon, communication, isolation, reopening or restoration. Official incident instructions always take precedence.

WHY THIS IS USEFUL

The incident structure can maintain a common verified impact picture while keeping command choices with authorised owners.

LB-064

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which alternative spaces, routes and service arrangements are currently usable for the affected activities, not merely listed on paper?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current room availability, environmental and power status, route and lift availability, network and teaching-technology heartbeat, and anonymous capacity context. Spatial coverage: candidate alternatives and the full journey or service chain needed to use them. Observation and aggregation: current event-driven status, recent physical-check timestamps and incident snapshots. Decision-use gate: current mapping, owner-confirmed function and capacity, all critical services healthy and a physical validation where telemetry cannot establish usability. Known limits: availability does not prove suitability, accessibility, security, specialist capability or user acceptance. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: incident command, space management, timetabling, accessibility, IT, Estates, research facilities, security and affected activity owners. Required records: official closures, current room inventory, capacities, bookings, access and safety assessments, service status, continuity plans and relocation constraints. Source precedence: incident instruction and current competent access or safety advice outrank room analytics; the May 2021 plan is historical only. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: alternative status is stale immediately after booking, closure, route, service or incident change. Unresolved conflicts: room function, availability, capacity, accessibility, service capability, priority and affected-owner acceptance; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: alternative-space and route feasibility matching. Decision interval: during disruption and in continuity planning. Minimum evidence: current requirement, verified candidate status, capacity, essential services, access route, security need and accountable receiving owner. Method: screen mandatory requirements, match remaining candidates, quantify capacity or service gaps and present unknowns requiring physical or professional check. Quantified uncertainty: show status age, unverified features, demand range and confidence in each alternative; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to incident command, space owner and affected activity owner.

ANSWER BOUNDARY

The service must not infer individual suitability, willingness, disability or activity priority from aggregate evidence. It must not disclose confidential bookings, research requirements, personal adjustments or security-sensitive layouts beyond authorised roles. It must not promise or certify safe or equivalent accommodation, universal accessibility or continuity of the activity; it must not authorise or control relocation, booking override, access, service setup or reopening. Affected owners and competent advisers must approve use.

WHY THIS IS USEFUL

Incident and space owners can select a currently usable alternative and avoid directing people to nominal but unsuitable provision.

LB-065

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which dependency-aware recovery order should incident leaders adopt to restore priority services safely with the people and resources currently available?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current service and asset status, dependency events, corrective work progress, environmental conditions, access routes and available backup capacity. Spatial coverage: the affected service graph and verified recovery resources, with restricted detail limited to authorised roles. Observation and aggregation: event-driven updates and scenario snapshots at command-approved cut-offs. Decision-use gate: current incident boundary, owner-validated dependencies, competent safety constraints and time-stamped corrective work estimates with confidence ranges. Known limits: corrective work duration, hidden damage, staff and supplier availability and common-cause failures remain uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: incident command, business continuity, Estates, IT, security, research facilities, procurement and priority-service owners. Required records: official incident objectives, restoration plans, dependencies, safety constraints, resource availability, supplier estimates and priority outcomes. Source precedence: incident command and competent technical sequence constraints outrank analytic optimisation. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: recovery evidence is stale at each new finding, resource change or command decision. Unresolved conflicts: mission priority, technical dependency, corrective work estimate, safety constraint and ownership of trade-offs; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: dependency-constrained recovery sequencing and scenario comparison. Decision interval: during recovery at command-approved cadence and in exercises. Minimum evidence: current priorities, verified dependencies, resource limits, safety constraints, corrective work ranges and owner for every restoration step. Method: generate feasible sequences, test delay and failure scenarios, expose bottlenecks and present trade-offs among mission value, risk and recovery time. Quantified uncertainty: show duration ranges, sequence sensitivity, unverified dependencies and share of tested cases in which the proposed order becomes infeasible; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to incident command and accountable technical and service owners.

ANSWER BOUNDARY

The service must not infer that the mathematically highest-value sequence is operationally acceptable. It must not disclose security-sensitive dependencies, personal availability or restricted research consequence beyond authorised roles. It must not promise or certify restoration time, safety, resource availability or mission priority; it must not authorise or control recovery order, dispatch, isolation, staffing, procurement or reopening. Command retains authority to change sequence as conditions evolve.

WHY THIS IS USEFUL

Recovery owners can agree a dependency-aware sequence, resource it and explain why lower-priority services wait.

LB-066

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How exposed are priority services to heat, flooding, severe weather, supply failure, transport disruption and external works?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: local temperature, humidity, rainfall or water-level points where justified, plant and utility status, leak sensors, external intake conditions and service availability. Spatial coverage: priority service chains and vulnerable locations verified against current assets and site conditions. Observation and aggregation: 5-15 minute local conditions, official forecast or warning updates and seasonal summaries. Decision-use gate: calibrated local sensors, current exposure mapping, authoritative external data and known coverage gaps. Known limits: local sensors do not predict hazards, future climate or off-site supply and transport conditions. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: risk and business-continuity owners, Estates, sustainability, security, procurement, transport and external utility or emergency authorities. Required records: current risk assessments, flood and weather information, asset vulnerability, supplier and transport dependencies, incident history and adaptation plans. Source precedence: official warnings, competent risk assessment and current on-site evidence outrank analytic scenarios or historical floor plans. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: exposure and vulnerability evidence is stale after asset, site, supplier, climate method or external infrastructure change. Unresolved conflicts: hazard data, local exposure, asset vulnerability, supplier dependency and scenario assumptions; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: multi-hazard exposure, vulnerability and consequence scenario analysis. Decision interval: seasonally, annually and when official warnings apply. Minimum evidence: authoritative hazard source, current asset exposure, vulnerability evidence, priority service and at least two plausible severity scenarios. Method: separate hazard, exposure, vulnerability and consequence; identify trigger points and common protections; do not convert scenarios into forecasts. Quantified uncertainty: show source resolution, scenario range, non-stationarity, unmodelled external dependency and confidence in local vulnerability; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable risk owner and relevant competent engineering or external authority.

ANSWER BOUNDARY

The service must not infer event probability, safety, future climate outcome or individual travel behaviour from scenario data. It must not disclose restricted vulnerabilities, personal travel information or supplier-sensitive details beyond authorised roles. It must not promise or certify forecast, regulatory adequacy, resilience or absence of risk; it must not authorise or control closure, evacuation, travel advice, investment, supplier action or risk acceptance. Official warnings and incident instructions take precedence.

WHY THIS IS USEFUL

Risk and continuity owners can set hazard-specific triggers, adaptations and contingencies for the services with greatest exposure.

LB-067

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Do redundancy, spare capacity, manual workarounds and supplier arrangements operate end to end under realistic test conditions?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: primary and backup service status, transfer events, load, runtime, environmental conditions and test-step completion. Spatial coverage: the full continuity chain from detection and decision through transfer, operation, resupply and restoration. Observation and aggregation: event-level exercise data, periodic heartbeat and post-test summaries. Decision-use gate: owner-approved test script, calibrated instrumentation, independent observation and reconciliation of expected and actual states. Known limits: scheduled tests may not reproduce incident stress, staff absence, common cause or supplier congestion. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: business continuity, technical service owners, procurement, suppliers, School operations, security and exercise governance. Required records: continuity plan, test script, contracts, competence, spares, load assumptions, observations, defects and corrective actions. Source precedence: observed end-to-end test evidence outranks design intent, inventory listing or supplier assurance. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: test evidence is stale after configuration, staffing, supplier, load, procedure or asset change. Unresolved conflicts: designed versus tested capability, workaround ownership, supplier commitment, load and unresolved test defects; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: end-to-end continuity test evaluation and gap analysis. Decision interval: after each exercise or material change. Minimum evidence: approved scenario, objective success criteria, complete event log, independent observations, current configuration and owner for every failed step. Method: compare expected and actual sequence, measure transfer and endurance, identify common-cause and human dependencies, and retest material corrections. Quantified uncertainty: show test realism, untested branches, observer limitations and confidence in transfer to a real incident; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the business-continuity lead and owners of failed technical, process or supplier steps.

ANSWER BOUNDARY

The service must not infer future incident performance, staff competence or supplier reliability from one exercise. It must not disclose personal performance data, security-sensitive procedures or commercially restricted supplier detail beyond authorised roles. It must not promise or certify resilience, contract performance, safe operation or adequate training; it must not authorise or control disciplinary action, supplier remedy, operational change, test frequency or investment. Exercises provide bounded evidence, not guarantees.

WHY THIS IS USEFUL

Owners can close gaps between documented fallback and tested performance, including suppliers, people, access and communications.

LB-068

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which lessons from previous disruptions remain unowned, overdue, untested or invalidated by later building changes?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: incident-relevant service events, restoration status, fault recurrence and current health of assets or controls named in corrective actions. Spatial coverage: only incidents and actions in the authorised lessons register, with sensitive detail restricted by role. Observation and aggregation: incident timelines, monthly action status and recurrence monitoring. Decision-use gate: owner-confirmed incident record, action identifier, due date, closure evidence and current mapping to the affected service. Known limits: absence of recurrence does not prove the action worked; organisational lessons may not be sensor-observable. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: incident owner, business continuity, governance, Estates, IT, security, School operations and action owners. Required records: post-incident assessments, lessons, action tracker, evidence of implementation, test results, current ownership and later related incidents. Source precedence: approved assessment findings and current action-owner evidence outrank inferred closure from telemetry. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: actions are stale after owner, scope, service or risk change and cannot remain closed without required validation. Unresolved conflicts: lesson wording, action ownership, closure evidence, test adequacy and recurrence classification; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: lesson-to-action traceability, overdue and effectiveness analysis. Decision interval: monthly and at resilience governance assessment. Minimum evidence: approved lesson, named action and owner, due date, implementation evidence, validation method and current service link. Method: classify open, implemented, validated, ineffective, superseded and unowned actions; link recurrence only where incident taxonomy and evidence support it. Quantified uncertainty: show missing validation, ownership gaps, small recurrence counts and confidence that an action addresses the original lesson; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the incident or continuity governance owner and accountable action owner.

ANSWER BOUNDARY

The service must not infer individual blame, negligence or competence from overdue actions or recurring incidents. It must not disclose personal incident evidence, restricted vulnerabilities or privileged assessment material beyond authorised roles. It must not promise or certify control effectiveness, absence of recurrence or closure of risk; it must not authorise or control disciplinary action, action closure, risk acceptance, testing or investment. A documented action is not a verified improvement.

WHY THIS IS USEFUL

Governance can restore ownership, retest actions and close or replace lessons that no longer match the current building.

LB-069

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which early indicator should trigger a readiness check, while being kept separate from any prediction that failure will occur?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: owner-approved trends in fault recurrence, alarm frequency, runtime, battery state, environmental drift, service load, supplier delay or backup-test results. Spatial coverage: only critical services with a defined failure mode and validated leading-indicator rationale. Observation and aggregation: service-appropriate samples, weekly trends and threshold events. Decision-use gate: stable measurement, sufficient baseline, known false-alarm rate and current linkage between indicator and the service concern. Known limits: many indicators drift without failure and some failures occur without warning. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: asset and service owners, risk, maintenance, business continuity, procurement and competent technical advisers. Required records: failure history, condition and maintenance records, risk assessment, supplier data, test results and owner-approved readiness thresholds. Source precedence: competent condition or risk assessment outranks statistical alerting; thresholds must have a named response owner. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: indicator evidence is stale after asset, duty, maintenance, supplier or failure-mode change. Unresolved conflicts: baseline, alert threshold, false alarms, failure classification and owner interpretation; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: leading-indicator validation and alert performance analysis. Decision interval: monthly and after each alert outcome. Minimum evidence: defined failure mode, stable baseline, enough alert and non-alert periods to estimate performance, current owner and proportionate response. Method: use control limits or survival models where justified, report sensitivity and false-positive rate, and frame the output as a check trigger. Quantified uncertainty: show event scarcity, model drift, confidence intervals and cases where predictive performance cannot be estimated; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable asset or service owner and risk or continuity lead.

ANSWER BOUNDARY

The service must not infer failure certainty, unsafe condition or staff or supplier performance from an alert. It must not disclose restricted vulnerabilities, personal work data or commercially sensitive supplier information beyond authorised roles. It must not promise or certify prediction, remaining life, resilience or need for shutdown; it must not authorise or control inspection, maintenance, shutdown, supplier action or incident response. The safe action is a proportionate readiness check, not automatic control.

WHY THIS IS USEFUL

Operations can define a proportionate readiness trigger and avoid presenting weak indicators as failure forecasts.

LB-070

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which resilience option gives the most robust reduction in service consequence and recovery time across several credible scenarios?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: baseline availability, incident duration, dependency, backup capacity, service-load and environmental telemetry relevant to candidate measures. Spatial coverage: critical service chains and affected scenarios, using a common boundary for each option. Observation and aggregation: historic incidents, test evidence and scenario summaries over the planning horizon. Decision-use gate: current dependency map, owner-validated consequence, tested baseline controls and explicit data-quality class for each scenario. Known limits: rare events, future hazards, common cause and behavioural adaptation make expected-loss estimates uncertain. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: risk and business-continuity owners, School and University leadership, Estates, IT, security, finance, procurement and affected services. Required records: business-impact analysis, risk register, incident history, test results, option cost, lifecycle, dependencies, benefits and residual risk. Source precedence: mandatory safety and current risk-owner judgement constrain options before cost-effectiveness comparison. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: scenario, cost and control evidence is stale after hazard, service, supplier, asset or policy change. Unresolved conflicts: consequence values, scenario likelihood, common benefits, residual risk, cost and ownership; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: multi-scenario resilience option appraisal and robustness analysis. Decision interval: annually and at investment gateways. Minimum evidence: defined priority services, credible scenarios, tested baseline, option mechanisms, cost ranges, residual risks and named benefit owners. Method: estimate consequence and recovery ranges by scenario, identify options with benefits across scenarios, test correlated failures and avoid false precision in event probability. Quantified uncertainty: show scenario ranges, value assumptions, rank reversals, unquantified benefits and conditions where an option does not work; report tested versus untested branches, scenario pass counts and the observed runtime or recovery range. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable risk owner, finance partner and technical service owners.

ANSWER BOUNDARY

The service must not infer precise event probability, guaranteed resilience or institutional risk appetite from the model. It must not disclose security-sensitive vulnerabilities, commercial options or personal dependencies beyond authorised roles. It must not promise or certify value for money, safety, compliance, deliverability or risk acceptability; it must not authorise or control investment, procurement, risk acceptance or service design. The appraisal informs accountable prioritisation; it does not approve risk.

WHY THIS IS USEFUL

Leadership can prefer measures that protect several services and scenarios, or identify where a specialist measure remains necessary.

LB-071

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Who is accountable for every reported outcome, source, threshold, exception and resulting action, and where is ownership missing?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: source-health, update, threshold and exception metadata for approved dashboards and building-service feeds. Spatial coverage: only metrics and evidence routes in the current governance pack. Observation and aggregation: continuous health events and a snapshot at each reporting cut-off. Decision-use gate: every metric has a named source owner, outcome owner, threshold owner, action owner and tested escalation route. Known limits: technical ownership does not establish decision authority or professional competence. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: governance secretary, School and University planning, Estates, data owners, risk, finance, sustainability and service owners. Required records: current responsibility matrix, metric dictionary, source register, committee terms, delegations, action tracker and escalation routes. Source precedence: formal current delegations and accountable-owner confirmation outrank inferred responsibility from system access or past practice. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: ownership is stale after organisational, delegation, committee, contract or service change. Unresolved conflicts: local and central responsibility, data custody, professional accountability, action ownership and delegated authority; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: responsibility and escalation-path completeness analysis. Decision interval: at each governance cycle and organisational change. Minimum evidence: defined metric or decision, current delegation, named owners for source, interpretation, decision and action, and an escalation route. Method: map RACI-like roles without collapsing accountability into data ownership, flag unowned or conflicting roles and require owner confirmation. Quantified uncertainty: show unresolved ownership, expired delegation and functions represented only by informal practice; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the governance secretary and accountable executive owner.

ANSWER BOUNDARY

The service must not infer authority, competence or accountability from job title, data access or previous action. It must not disclose personal performance information or restricted governance material beyond authorised roles. It must not promise or certify valid delegation, legal accountability or adequacy of governance; it must not authorise or control appointment, delegation, escalation, action ownership or committee decision. Formal governance records remain authoritative.

WHY THIS IS USEFUL

Governance can fill ownership gaps, set escalation dates and ensure every reported exception has a responsible decision route.

LB-072

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are leadership reports using current definitions, boundaries, time windows and source versions so that figures are genuinely comparable?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: metric lineage, source-health, mapping version, coverage, timestamps and restatement flags for every reported sensor-derived value. Spatial coverage: the exact measures, assets, spaces and periods in the report. Observation and aggregation: source-level updates plus a frozen reporting snapshot at the decision cut-off. Decision-use gate: metric dictionary match, current mappings, owner-approved transformations and completeness above the threshold stated for each measure. Known limits: a technically fresh number can remain conceptually incomparable or outside scope. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: planning and performance, governance, finance, Estates, sustainability, data owners and relevant service owners. Required records: metric dictionary, reporting boundary, source register, approved restatements, prior reports, current schedules and owner commentary. Source precedence: current owner-approved definitions and formal restatements outrank continuity with an obsolete series. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: figures are stale after source, boundary, definition, period, mapping or material operational change. Unresolved conflicts: definitions, denominators, floor area, service hours, period cut-offs, restatements and local versus central versions; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: report comparability, lineage and freshness validation. Decision interval: before each leadership pack is issued. Minimum evidence: metric definition and owner, boundary, period, cut-off, source lineage, coverage, restatement status and prior comparable value. Method: validate each figure against the dictionary, block unsupported comparisons, annotate restatements and separate new-series baselines from trends. Quantified uncertainty: show completeness, definition change, unavailable comparator and confidence in comparability; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the report owner and each source or metric owner.

ANSWER BOUNDARY

The service must not infer performance change from incomparable or restated numbers. It must not disclose restricted source detail, personal data or small organisational units beyond authorised roles. It must not promise or certify accuracy, completeness or comparability where lineage or definitions are unresolved; it must not authorise or control report approval, target, action, sanction or public disclosure. Unknown or non-comparable must not be displayed as zero or trend.

WHY THIS IS USEFUL

Leadership receives comparable figures with visible definition, boundary and version changes instead of false trends.

LB-073

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which performance indicators risk rewarding substitution, gaming or under-reporting rather than the intended building outcome?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: metric-specific aggregate service, demand, quality, fault, energy, access or response data, plus denominator and missingness metadata. Spatial coverage: the complete service boundary affected by the indicator, including likely displaced work and downstream effects. Observation and aggregation: operational data at source resolution and monthly or term reporting. Decision-use gate: stable definitions, complete denominator, independent check measure and enough history to inspect threshold effects. Known limits: unmeasured quality, low-volume severe events and activity shifted outside the metric can remain invisible. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: performance and planning, service owner, finance, procurement, risk, accessibility, sustainability and stakeholder-outcome owners. Required records: approved outcome map, metric dictionary, service standards, contract incentives, assurance check results, complaints and known unintended consequences. Source precedence: mandatory duties and outcome-owner interpretation outrank target attainment; independent evidence outranks self-reported metric alone. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: an indicator is stale after service, incentive, denominator or outcome-definition change. Unresolved conflicts: target behaviour, denominator exclusions, quality, displaced cost, severe exceptions and stakeholder outcome; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: indicator validity, balance and gaming-risk assessment. Decision interval: annually and before target or contract renewal. Minimum evidence: defined outcome, causal rationale, denominator, complementary quality or harm measure, history across threshold periods and named owner. Method: test construct validity, discontinuities near targets, substitution and missing denominator; use balanced indicators where one measure can be gamed. Quantified uncertainty: show measurement error, unobserved harm, target sensitivity and whether the indicator-outcome relationship remains supported; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the outcome owner, metric owner and relevant risk or contract owner.

ANSWER BOUNDARY

The service must not infer staff, contractor or unit intent or performance from metric patterns. It must not disclose personal performance, small-team results or commercially restricted contract evidence beyond authorised roles. It must not promise or certify true outcome achievement, absence of gaming or fairness of an incentive; it must not authorise or control targets, sanctions, payment, staffing or contract design. A metric is evidence about a defined construct, not the construct itself.

WHY THIS IS USEFUL

Measure owners can replace or balance indicators that drive harmful substitution, gaming or hidden unmet demand.

LB-074

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Can each material building decision be reconstructed from the evidence, assumptions, permissions, advice and uncertainty available at the cut-off?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: immutable source references, metric snapshots, health, coverage, mapping and timestamp metadata for evidence cited in the decision. Spatial coverage: only the decision scope and evidence pack, with access controlled to the originating permissions. Observation and aggregation: a frozen snapshot at the decision cut-off plus later amendments kept as separate versions. Decision-use gate: complete source identifiers, owner, version, effective-observed-retrieved times, transformations and access decision. Known limits: traceability cannot prove the decision was correct or that all relevant evidence was known. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: decision owner, governance, records management, legal, finance, risk, Estates and all material evidence or advice owners. Required records: decision paper, minutes, approvals, evidence register, assumptions, professional advice, permissions, dissent and subsequent updates. Source precedence: the contemporaneous approved record outranks reconstructed narrative; later evidence must not be backdated into the decision basis. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: the trace is incomplete if versions, cut-off, permissions or material advice are missing. Unresolved conflicts: decision versions, evidence ownership, assumptions, dissent, advice scope and later amendments; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: decision-lineage and contemporaneous evidence reconstruction. Decision interval: at each material decision and periodic governance assessment. Minimum evidence: decision identifier, authority, options, cut-off, frozen evidence pack, assumptions, advice, permissions, dissent and recorded rationale. Method: build a trace from each material conclusion to its source, distinguish contemporaneous and later evidence, and flag unsupported or inaccessible premises. Quantified uncertainty: show trace completeness, unresolved source conflicts and evidence that existed but was outside permission or not available then; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the governance secretary and accountable decision owner.

ANSWER BOUNDARY

The service must not infer decision quality, legality, good faith or individual responsibility from the assurance check trail. It must not disclose privileged advice, personal data, security-sensitive evidence or commercial material beyond authorised roles. It must not promise or certify correctness, completeness, compliance or defensibility of the decision; it must not authorise or control assessment outcome, accountability finding, approval, sanction or disclosure. Traceability preserves evidence and judgement; it does not replace governance assessment.

WHY THIS IS USEFUL

The decision owner can demonstrate what was known, permitted and uncertain at the time without rewriting the historical record.

LB-075

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Where do authoritative records or competent professional assessments conflict, and who has the mandate to resolve each conflict?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: source status, mapping, timestamps and evidence lineage for the specific telemetry or register entries in disagreement. Spatial coverage: only the disputed claim and affected asset, space, service or period. Observation and aggregation: current source snapshots and a preserved history of changes. Decision-use gate: identified owners, scopes, effective dates, versions and no automatic overwrite of one source by another. Known limits: technical recency does not determine authority and some disagreements reflect different valid scopes. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: source owners, relevant duty holder or professional lead, governance, records management and accountable decision owner. Required records: authoritative records, signed assessments, scope statements, delegations, change history, assumptions and conflict log. Source precedence: apply approved precedence only within each source's competence and scope; preserve material dissent for the authorised resolver. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: evidence is stale under its own owner-defined rule, not because another source is newer. Unresolved conflicts: asset identity, spatial boundary, effective date, professional scope, legal interpretation and organisational authority; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: source-conflict classification and resolution routing. Decision interval: when the conflict becomes material and at decision cut-off. Minimum evidence: precise disputed claim, each source and owner, scope, effective date, authority, consequence and named resolver or escalation route. Method: classify factual, temporal, definitional, scope and professional-judgement conflicts; reconcile only where an authorised rule exists. Quantified uncertainty: state unresolved positions, evidence strength and how each interpretation changes the decision; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable resolver named in current governance or the duty holder where professional authority applies.

ANSWER BOUNDARY

The service must not infer that a majority, average or newest source is correct. It must not disclose privileged advice, personal information or restricted professional material beyond authorised roles. It must not promise or certify resolution, legal interpretation, professional opinion or decision correctness; it must not authorise or control source amendment, acceptance of advice, escalation or decision. Material conflict must remain visible until an authorised owner resolves it.

WHY THIS IS USEFUL

Governance can route each conflict to the authorised resolver and keep competing evidence visible until a decision is recorded.

LB-076

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How do plausible demand, cost, climate, policy, technology and service scenarios alter the preferred option or trigger point?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: current aggregate demand, energy, service-load, asset-condition and reliability baselines relevant to scenario variables. Spatial coverage: common option boundaries and the services materially affected by each scenario. Observation and aggregation: historic baseline plus scenario values at named planning horizons. Decision-use gate: verified baseline, owner-approved scenario ranges, documented correlations and explicit non-sensor assumptions. Known limits: scenarios are conditional and cannot predict policy, technology adoption, climate events or stakeholder behaviour. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: University planning, finance, Estates strategy, School leadership, sustainability, risk, policy and technology owners. Required records: approved planning assumptions, cost bases, climate scenarios, policy requirements, technology roadmaps, service plans and option models. Source precedence: current approved scenario sources and competent professional assumptions outrank ad hoc forecasts. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: scenario inputs are stale after policy, market, service, technology, climate-method or planning change. Unresolved conflicts: scenario definitions, correlations, planning horizons, costs, service assumptions and owner judgements; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: multi-scenario option robustness and trigger analysis. Decision interval: at strategy refresh and each major gateway. Minimum evidence: common option set, approved scenario ranges, baseline, planning horizon, correlations, constraints and owner for each key assumption. Method: evaluate each option across coherent scenarios, identify rank reversals and threshold values, and distinguish robust, contingent and dominated options. Quantified uncertainty: show uncertainty bands, scenario dependence, unmodelled shocks and the assumptions with greatest decision leverage; report low, central and high cases and the share of plausible parameter sets that change the preferred action. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the planning owner and accountable option or risk owners.

ANSWER BOUNDARY

The service must not infer future demand, policy, cost, climate or technology outcome from scenarios. It must not disclose commercially sensitive forecasts, restricted vulnerabilities or personal planning data beyond authorised roles. It must not promise or certify forecast accuracy, affordability, value, safety or inevitability of an option; it must not authorise or control strategy, target, investment, procurement or trigger activation. Scenarios support preparedness, not prediction.

WHY THIS IS USEFUL

Leadership can choose a robust option, define scenario triggers or defer where the preferred choice is too assumption-sensitive.

LB-077

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Which commitments, actions and dependencies are overdue, blocked, duplicated or at risk of losing an accountable owner?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: action-status, dependency, due-date and evidence-link metadata from approved trackers, plus current service evidence where an action claims completion. Spatial coverage: building-related commitments in the current governance scope. Observation and aggregation: daily status updates and frozen snapshots at committee cut-offs. Decision-use gate: unique identifiers, named owner, current due date, dependency owner, closure evidence and duplicate detection. Known limits: administrative status can lag work, conceal weak closure evidence or omit informal dependencies. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: governance secretary, programme and project owners, Estates, risk, sustainability, accessibility, finance and service owners. Required records: approved action logs, committee decisions, project plans, risk treatments, benefit commitments, dependencies and closure evidence. Source precedence: current formal commitment and accountable-owner evidence outrank inferred progress from system activity. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: status is stale after due-date, owner, scope, dependency or decision change. Unresolved conflicts: duplicate actions, conflicting due dates, owner changes, dependency status and sufficiency of closure evidence; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: commitment dependency, ownership and closure-quality analysis. Decision interval: before each governance meeting. Minimum evidence: current action identifier, decision source, owner, due date, dependencies, status, evidence and escalation rule. Method: identify overdue and blocked paths, cluster genuine duplicates without merging distinct obligations, and distinguish completed work from validated outcome. Quantified uncertainty: show missing updates, uncertain duplicates, critical-path sensitivity and actions with no current accountable owner; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the governance secretary and accountable programme, project or action owner.

ANSWER BOUNDARY

The service must not infer individual performance, negligence or intent from overdue or blocked status. It must not disclose personal work detail, privileged committee material or restricted project information beyond authorised roles. It must not promise or certify completion, outcome achievement or adequacy of the action; it must not authorise or control reassignment, escalation, closure, sanction or programme decision. Administrative closure requires traceable evidence appropriate to the commitment.

WHY THIS IS USEFUL

Programme owners can reassign, unblock, combine or retire commitments before dependencies become orphaned.

LB-078

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

Are data access, linkage, retention and disclosure no greater than needed for each approved leadership purpose?**SENSORS AND TELEMETRY**

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: data-source identity, aggregation level, access, query, export, retention and deletion assurance check metadata - not the underlying personal content. Spatial coverage: only data routes used by the Talking Abacws leadership service. Observation and aggregation: event logs with periodic access assessment and decision-specific snapshots. Decision-use gate: complete source catalogue, purpose, lawful or institutional basis, permission owner, aggregation rule and tested deletion or expiry path. Known limits: assurance check logs may not capture off-system copies, onward disclosure or whether a use remains ethically proportionate. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: information governance, data protection, cybersecurity, records management, source owners, ethics or research governance where applicable, and decision owners. Required records: data inventory, permissions, agreements, privacy assessments, retention schedules, access roles, disclosure decisions, incidents and assessment dates. Source precedence: current legal and institutional governance decisions and source-owner restrictions outrank analytic convenience. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: permission is stale at expiry or after purpose, data, linkage, recipient, system or risk change. Unresolved conflicts: purpose, lawful basis, owner permission, aggregation, retention, onward use and security classification; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: purpose-to-data minimisation, permission and retention compliance checking. Decision interval: before access, at material change and periodic assessment. Minimum evidence: defined decision purpose, necessary fields, minimum aggregation, recipients, owner permission, retention period and risk assessment. Method: test necessity and proportionality, remove unused fields and linkages, suppress small groups, verify access and deletion, and block unapproved secondary use. Quantified uncertainty: state unverified copies, ambiguous purpose, residual re-identification risk and controls not technically observable; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the information-governance and source owners with the accountable decision owner.

ANSWER BOUNDARY

The service must not infer consent, legality, fairness or absence of privacy risk from technical controls alone. It must not disclose personal data, raw credentials, research content or sensitive cases beyond approved need. It must not promise or certify data-protection compliance, anonymity, security or ethical acceptability; it must not authorise or control access approval, retention, disclosure, linkage or incident response. No leadership convenience justifies expanded data use.

WHY THIS IS USEFUL

Information governance can remove unnecessary data, narrow permissions and set proportionate retention and disclosure controls.

LB-079

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

How can buildings or organisational units be compared without false precision, unfair ranking or incentives to suppress legitimate demand?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: common aggregate service, demand, energy, cost, reliability, access or environmental indicators with coverage and data-quality metadata. Spatial coverage: only units meeting an approved comparison boundary and safe aggregation threshold. Observation and aggregation: common annual or term periods with contextual profiles. Decision-use gate: shared definitions, matched periods, known floor area and service intensity, minimum data quality and no unsupported composite score. Known limits: mission, building form, specialist loads, hours, climate, investment history and data quality limit comparability. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: planning, Estates, finance, sustainability, accessibility, service owners and the units being compared. Required records: current metric definitions, peer-group rationale, floor area, service hours, mission context, cost bases, asset condition and restatement history. Source precedence: approved like-for-like definitions and competent contextual interpretation outrank simple rank order. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: benchmark evidence is stale after definition, boundary, period, restatement or material service change. Unresolved conflicts: peer selection, normalisation, floor area, cost base, specialist activity, context and public or internal use; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: context-adjusted distribution and peer-learning analysis. Decision interval: annually. Minimum evidence: common definitions and period, justified peer group, minimum data quality, material context variables and permission to compare. Method: show distributions and ranges, normalise only defensible variables, identify outliers for local inquiry and avoid collapsing conflicting dimensions into one rank. Quantified uncertainty: show residual non-comparability, sample size, model error and sensitivity to peer-group choice; report valid-data coverage %, numerator and denominator, and a 95% interval or stated measurement tolerance where valid. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the planning and metric owners with representatives of compared units.

ANSWER BOUNDARY

The service must not infer quality, efficiency, staff performance or institutional merit from rank or residual. It must not disclose small-unit data, restricted operations or results outside the agreed comparison purpose. It must not promise or certify fairness, causation, transferability or a definitive league table; it must not authorise or control targets, sanctions, resource allocation, publication or performance management. Comparison should support inquiry and peer learning, not judgement by rank.

WHY THIS IS USEFUL

Leaders can compare service context and outcomes fairly, identify peers for learning and avoid punitive league tables.

LB-080

SCHOOL LEADERSHIP, UNIVERSITY LEADERSHIP, AND BUILDING OWNERS - STAKEHOLDER QUESTION

What minimum evidence pack should leadership require before approving, deferring or escalating a material building matter?

SENSORS AND TELEMETRY

Proposed evidence only; use is conditional on independently verified deployment, current asset and location mapping, commissioning, permission and live health: only verified aggregate observations material to the decision, with source health, mapping, coverage, timestamp, quality and transformation metadata. Spatial coverage: the exact assets, spaces, services, people-facing outcomes and periods within the decision scope. Observation and aggregation: captured to a declared decision cut-off, with authorised historical snapshots preserved as separate versions. Decision-use gate: every cited observation has current mapping, owner, permission, completeness and a clear relationship to a material decision claim. Known limits: too little evidence hides uncertainty while excess detail can obscure judgement and increase disclosure risk. Privacy output: aggregates only; no identities, images, audio content, raw device identifiers or personal journey reconstruction.

AUTHORITATIVE SUPPORTING DATA

Accountable source owners: the accountable decision owner, governance, finance, risk, safety, accessibility, sustainability, legal and service owners as relevant. Required records: decision and authority, options, objectives, constraints, current evidence, advice, provenance, permissions, stakeholder effects, dissent and next gateway. Source precedence: mandatory duties, current competent advice and authoritative records precede analytic summaries; material disagreement remains visible. The May 2021 floor plan is historical orientation only and cannot establish current use, access, services, hazards or closures. For every item retain owner, permission basis, provenance and version, transformations, effective time, observed time, retrieved time and the frozen decision cut-off. Staleness: the pack is stale after new material evidence, changed option, advice, cost, risk or decision cut-off. Unresolved conflicts: option versions, missing owners, stale advice, incompatible evidence, permissions and uncertainty that could change the decision; keep competing positions visible and route resolution to the authorised owner.

ANALYSIS REQUIRED

Operation type: decision-focused evidence synthesis and sufficiency checking. Decision interval: for every material building decision. Minimum evidence: decision and authority, options including no-change, objectives, non-negotiable constraints, evidence lineage, uncertainty, affected stakeholders, dissent and safe-failure route. Method: produce an answer-first summary, claim-evidence table, option ranges, dependencies, open gates and traceable supporting detail; distinguish fact, estimate, scenario and judgement. Quantified uncertainty: show residual uncertainty, evidence grade, missing voices, unresolved conflict and which additional evidence could change the choice; report source completeness %, age of the oldest cited item and the number of unresolved conflicts or open actions. Safe failure: if mapping, permission, freshness, health, minimum coverage or material source agreement is absent, return "insufficient evidence", name the missing condition and route it to the accountable decision owner and governance secretary.

ANSWER BOUNDARY

The service must not infer that a concise pack is complete or that analytic balance determines the institutional choice. It must not disclose personal, privileged, security-sensitive or commercially restricted material beyond authorised roles. It must not promise or certify legality, compliance, safety, affordability, value or success; it must not authorise or control approval, deferral, escalation, funding, communication or operational control. The pack must preserve accountable human judgement and professional advice.

WHY THIS IS USEFUL

Leadership gains a repeatable evidence gate that supports approval, deferral or escalation without transferring accountability to analytics.

Catalogue summary

Eighty questions organised for accountable leadership decisions, not automated authority.

10

Mission, strategy and stakeholder outcomes

LB-001 - LB-010

10

Capacity, access, experience and service demand

LB-011 - LB-020

10

Operations, teaching, research and workplace performance

LB-021 - LB-030

10

Risk, safety, compliance and assurance

LB-031 - LB-040

10

Finance, assets, projects and long-term value

LB-041 - LB-050

10

Energy, carbon, inclusion and social value

LB-051 - LB-060

10

Resilience, disruption and business continuity

LB-061 - LB-070

10

Governance, accountability, scenarios and performance reporting

LB-071 - LB-080

Permitted answer states

Evidence-supported

Minimum sources agree within scope and uncertainty is explicit.

Conditional

Useful only if named assumptions, controls or time limits hold.

Insufficient evidence

A required source, permission, mapping, health check or minimum sample is absent.

Reserved decision

Evidence can be framed, but the choice remains with an authorised owner or competent professional.

Before acting

01

Confirm the accountable decision owner and competent advisers.

02

Freeze the evidence cut-off and record all three times.

03

Keep source conflicts, blind spots and suppressed cells visible.

04

Separate observed fact, estimate, scenario and judgement.

05

Name the next action, owner, due point and safe fallback.

The service supports accountable decisions; it does not replace accountable decision makers.

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